Do Shareholders Benefit from Green Bonds?*

Dragon Yongjun Tang
The University of Hong Kong
E-mail: yjtang@hku.hk

Yupu Zhang
The University of Hong Kong
E-mail: yupu@hku.hk

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Abstract

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Keywords: Green Bonds, Corporate Social Responsibility, Environmental, Social, and Governance (ESG), Institutional ownership, Stock liquidity, Investor attention.
JEL Classifications: G14; G32; G38; M14

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1. Introduction

Environmental, social, and governance (ESG) considerations are increasingly incorporated into corporate policies.[1] Hart and Zingales (2017) appeal for the maximization of shareholder welfare instead of stock returns. Magill, Quinzii, and Rochet (2015) provide a stakeholder model to show that when managers maximize the total value of the firm rather than that of shareholders only, a more sustainable equilibrium is achieved. However, empirical evidence on whether firms can benefit from improving the non-financial aspects of corporate activities, including ESG, is still lacking. In this paper, we fill the gap in the literature by empirically examining the nascent but fast-growing green bond market.

Green bonds are newly developed financial instruments with the specific goals of improving environmental impacts and social welfare. Similar to traditional fixed income securities, firms can issue green bonds to raise capital to finance their valuable investments. However, green bonds are intended to have a positive environmental benefit such as reducing CO_2 emissions and preventing pollution. Moreover, green bonds are certified by third parties. For first-time issuers, the process of issuing green bonds can be cumbersome and costly. Therefore, it is important to ask whether existing shareholders of the issuing firms can benefit from the issuance of green bonds. The answer to this question is useful for the further development of green bond market and impact investing.

We are among the first to provide a comprehensive empirical study on green bonds. In spite of the recent boom in the green bond market, there is still no universal definition of green bonds.[2] Broadly speaking, green bonds are fixed income securities issued by capital raising entities to fund their environmentally friendly projects, such as renewable energy, sustainable water management, pollution prevention, climate change adaptation and so on. Although the proceeds from green bonds are earmarked for green projects, the bonds are endorsed by the issuer's whole balance sheet. We compile the most comprehensive international green bond dataset for our empirical analysis. We start our data construction from the Climate Bonds Initiative (hereafter, CBI). CBI is a not-for-profit international organization with the mission of promoting capital solutions via green bonds for environment-related projects. CBI plays a prominent role in the green bond market and its

[1] See, e.g., Renneboog, Horst, and Zhang (2008), Benabou and Tirole (2006, 2010), Eccles, Ioannou and Serafeim (2014), Flammer and Kacperczyk (2016), Cao, Liang and Zhan (2018).
[2] See discussion in https://www.ft.com/content/ef9a02d6-28fe-11e7-bc4b-5528796fe35c

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Climate Bond Standard is adopted by many countries. Moreover, CBI approves the qualifications of third-party green bond verifiers such as KPMG, Deloitte, and Sustainalytics.[3]

Bloomberg has also launched initiatives and compiled data on green bonds. We augment the CBI dataset with Bloomberg's labeled green bond data.[4] There are several advantages to combining CBI with Bloomberg to build a comprehensive green bonds dataset. First, they have closely related and consistent definitions of green bonds. Thus, we can cross-validate the accuracy of our data. Second, Bloomberg provides adequate information about corporate bonds, including announcement date, issue amount, coupon, and maturity, that can be missing from the CBI data. Such data are important for our event study analysis.

We find from this dataset that the stock prices of issuers react positively to green bond issuance announcements. Specifically, there is a 1.4% cumulative abnormal return for the [-10, +10], 21-day event window around green bond issuance announcements. Moreover, the announcement returns are larger for first-time issuers than for repeat issuers. This announcement effect is both statistically significant and economically meaningful. Our results are consistent with prior literature on the relationship between corporate environmental performance and firm value, especially research examining the information content of corporate news on environmental issues using event studies.[5] Thus, issuing green bonds can improve firm value in the short run.

We explore three potential sources for this positive green bond announcement return: (1) The "*financing cost*" channel: socially responsible funds or investors with a green mandate may seek to hold green bonds to boost their ESG scores. As a result, those investors could push up the price and green bond issuers can benefit from a lower cost of debt, leading to a positive reaction by the stock market. (2) The "*investor attention*" channel: when firms label their green bonds, increasing media exposure can attract investors' attention and the visibility of the issuing firms can potentially increase, leading to more demand for their shares and a larger investor base. (3) The "*firm fundamental*" channel: green bonds demonstrate the firm's dedication to sustainable development,

[3] https://www.climatebonds.net/certification/approved-verifiers

[4] CBI is working with Bloomberg to enrich the green bond database. Their data coverage is slightly different. For example, Bloomberg requires 100% use of proceeds to be aligned with green activities.

[5] For example, Klassen and McLaughlin (1996) study when firms receive environmental awards, the stock market will react significantly positively.

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and investing in such projects can be valuable to firms in the long run, including by helping them survive adverse situations.

We examine green bond yield, institutional ownership, and stock market liquidity to understand the channels and mechanisms underlying positive announcement effects. In a broad sample, we find a green premium, that is, green bonds are issued at a yield spread that is 6.94 basis points lower than corporate bonds issued by similar firms. However, if we compare yield spread within the same issuing firm in the same year, we do not find any significant pricing difference. Overall, we do not have strong evidence supporting the "financing cost" channel. In other words, the positive stock market reaction is beyond the direct benefits of the lower cost of debt, if any, to the issuers.

When investigating the institutional ownership after green bond issuance compared with firms that issue conventional corporate bonds, we find a 7.9% increase in the institutional ownership (mostly driven by investment advisers and pension funds) of the green bond issuers. Interestingly, we find that hedge funds reduce their holdings in those firms after green bond issuance. Furthermore, the increase in institutional ownership comes mainly from domestic institutions. Such evidence is supportive of the "investor attention" channel of green bond effects.

We find that stock turnover increases significantly in relation to green bond issuance. Google search volume also spikes around event days. Such evidence suggests that the market does indeed pay attention to a firm's green bond development. Moreover, we find a significant improvement in stock liquidity in terms of bid-ask spreads and the Amihud measure after green bond issuance compared with matching firms issuing conventional corporate bonds. This result suggests that more media attention attract more investors and generate more trades on the stock market. It supports "investor attention" channel and against the "fundamental channel" because if investors believe the long term high valuation of the firm, they will realize the gain rather than hold the stock such that stock liquidity will shrink or hold still.

Together with several other contemporary studies, such as Baker, Bergstresser, Serafeim and Wurgler (2018), Flammer (2018), Reboredo (2018), and Febi, Schafer, Stephan and Sun (2018), our paper provides first evidence on the impact of green bond issuance on shareholders. Our findings contribute to the literature analyzing the effect of corporate social responsibility (CSR) or

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ESG on firm performance and firm value.[6] Margolis, Elfenbein, and Walsh (2007) find a positive but weak relationship between corporate social performance and corporate financial performance revealed in 167 studies over the past 35 years. However, the prior literature does not draw any conclusions about the positive effect of conducting CSR and environmental issues. There are also concerns that socially responsible firms suffer more from agency problems and that managers extract private benefits at the expense of existing shareholders (see, e.g., Jensen (2001) and Cheng, Hong, and Shue (2013)).[7]

An emerging strand of literature related to our study analyzes institutional investors and CSR or ESG (e.g., Dyck, Lins, Roth and Wagner (2018) and Chen, Dong and Lin (2018)). Dimson, Karakas, and Li (2015) show that active institutional investors engaging in environmental or social issues benefit firm from improved accounting performance and increased institutional ownership. Hoepner, Oikonomou, Sautner, Starks and Zhou (2018) examine institutional investor engagement in ESG issues and find that such engagements can benefit the firm by reducing the downside risk. Additionally, investors with a longer investment horizon prefer to hold high ESG firms and behave more patiently when incurring loss (Starks, Venkat and Zhu (2018)). Our findings are consistent with the above studies that institutional investors pay attention to and value firms' ESG performance.

Our study on green bonds can be useful for impact investing. Impact funds can hold green bonds to enhance their impact metrics, which measure their significant influence on the environment and society. The United Nations Principles for Responsible Investment (UNPRI, 2017) lists more than 1200 organizations representing $70 trillion assets under management as their signatories committing to socially responsible and sustainable investments.[8] Bialkowski and Starks (2016) suggest that SRI fund flow is different from conventional funds in terms of higher growth and more persistence. Geczy, Jeffers, Musto and Tucker (2018) find that the PE/VC industry adopts those impact investing features and addresses agency problems by contracting.

[6] The broad literature includes Sharfman and Fernando (2008), Hong and Kacperczyk (2009), Edmans (2011), Ghoul, Guedhami, Kwok and Mishra (2011), Goss and Roberts (2011), Hong and Kostovetsky (2012), Servaes and Tamayo (2013), Chava (2014), Bhandari and Javakhadze (2017), and Edmans, Li, and Zhang (2017).

[7] Cheng, Hong, and Shue (2013) find that increasing managerial incentives is associated with firm goodness, indicating that dollars spent on goodness are a result of agency problems.

[8] https://www.unpri.org/pri/about-the-pri

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Moreover, Barber, Morse and Yasuda (2018) show that impact funds derive utility from non-pecuniary investments while earning a lower rate of return.

Our study of green bonds also adds to the media attention literature. Grullon, Kanatas, and Weston (2004) show that the firm's visibility to investors has significant consequences for the stock market. In particular, firms that increase their capital expenditure on advertisement tend to have a larger number of institutional investors. Similarly, Frieder and Subrahmanyam (2005) find that individual investors prefer to invest in stocks with easily recognized products. Ben-Rephael, Da and Israelsen (2017) documents that post-earnings announcement drifts are driven by investor inattention to announcements and that institutional attention responds more quickly to major news events, leading retail attention. We also show that investors pay attention to the firm's green activities.

The remainder of this paper is organized as follows. Section 2 describes the background and presents the hypothesis development. Section 3 describes the data construction. Section 4 presents the results from event studies. Section 5 compares yield spread, institutional ownership and liquidity changes after issuing a green bond. Section 6 concludes the paper.

2. Background and Hypothesis Development

2.1. Green Bond Market Overview

The first green bond was called a "climate awareness bond" and was issued by the European Investment Bank (EIB) in 2007. Since then, the green bond market has continued to rise not only in total issue amount but also in other respects. For instance, the geographic base is expanding.[9] Moreover, bond types are also broadening. Poland became the first country to issue a green sovereign bond in December 2016, followed by France in January 2017. The United States government agency Fannie Mae issued the largest up-to-date $24.9 billion USD green mortgage-backed securities (MBS). In June 2017, Malaysia launched the world's first green Islamic bond, "green Sukuk," to finance climate-resilient growth.[10] Furthermore, issuer types are also enlarging meaningfully, including supranational organizations (i.e., World Bank and IFC), development

[9] Although the US, China and France lead the market, issuance is broadening, with first-time issuers from Argentina, Singapore, Chile, Fiji, Lithuania, Malaysia, Nigeria, the United Arab Emirates, Slovenia and Switzerland in 2017.
[10] https://www.worldbank.org/en/news/infographic/2017/09/19/malaysia-green-sukuk

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banks (i.e., ADB and AfDB), commercial banks (i.e., Bank of America and HSBC), non-bank financial institutions (i.e., REITS such as Regency Centers and Link) and corporations (i.e., Apple, Inc., and Tesla). According to Thomson Reuters, the annual issuance amount of green bonds reached $155.5 billion in 2017 and gained momentum to meet the $1 trillion target in 2020.[11]

Despite the fact that green bond market is growing rapidly, there are still disputes regarding definitions, guidelines and green taxonomy. In general, there are two green bond "standards": Green Bond Principles (GBP) and the Climate Bond Initiative (CBI).[12] GBP is a voluntary guideline established in 2014 by investment banks, including Bank of America Merrill Lynch, Citi, JPMorgan, BNP Paribas, and HSBC.[13] GBP requires transparency and disclosure and its four green bond components (use of proceeds, process of project evaluation and selection, management of proceeds, and reporting) are widely accepted by the market. Regardless of the generality of GBP, CBI provides eligible criterion and a detailed green taxonomy by sector that third parties can adopt to assess the qualification of a green bond. Therefore, we select the CBI database as a baseline to construct our international green bond database.

Our main focus in the paper is public listed green bond issuers. In our dataset, the first public green bond issuer is Nedbank, one of the largest banks in South Africa. Nedbank issued 4 billion ZAR in 2012 from a green savings bond program, which was used to finance renewable energy projects in South Africa.[14] The public green bond issuers were followed by Regency Centers Corp, a real estate investment trust (REIT), and Credit Agricole CIB, a diversified banking institution. Financial institutions and corporations are the major constituents among all the public green bond issuers. However, the designs of their green bonds are different: corporate green bonds are used to finance the issuers' own projects and as shown in Internet Appendix IA.1, project eligibility criteria are clearly specified; in contrast, financial institutions, including commercial banks, diversified banking institutions, investment companies and insurance companies, issue green bonds to make green loans to and invest in other firms to finance other firms' projects and as shown in Internet Appendix IA.2, they will only define general criteria to select green projects.

[11] See, e.g., https://www.reuters.com/article/greenbonds-issuance/global-green-bond-issuance-hit-record-155-5-billion-in-2017-data-idUSL8N1P5335
[12] https://www.bis.org/publ/qtrpdf/r_qt1709h.pdf
[13] https://www.climatebonds.net/market/best-practice-guidelines
[14] https://www.climatebonds.net/2014/05/sas-nedbank-490m-retail-green-bonds-offer

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From the issuers' perspective, green bonds can expand the breadth of ownership, enlarge their investor base and potentially obtain a lower cost of capital and longer tenor compared with straight corporate bonds. For the investors, green bonds may satisfy their green mandate and boost their ESG score. However, green bond issuers suffer from more information disclosures, upfront costs for certification and personnel, and reputation risk. Furthermore, there are no unified green bond standards to identify a green bond and limited enforcement of the law for supervising green integrity. Therefore, it is not a straightforward decision for corporations to issue green bonds.

Critics say that this fragmentation of standards and labeling creates uncertainty for investors and will slow the future growth of the market. However, a uniform standard to assess greenness, for example, a widely accepted criterion to evaluate green bonds, is difficult to implement. Furthermore, due diligence on environmental impacts is crucial to debt financing.[15] In other words, investors know little about the environmental impact of green loans and green bonds and the only information they can obtain is through voluntary disclosure or third-party verification. Additionally, all the above discussions enlighten us on the firm's incentive to issue green bonds and their real impact on firm performance after issuance.

2.2. Hypothesis Development

While there can be benefits from green bond issuance, there are also potential costs. One such cost is the fees paid to third-party certifiers and auditors to perform annual reviews. Indeed, the governments of Hong Kong and Singapore provide subsidies to issuers to cover such costs to encourage firms to issue green bonds in their domiciles. Therefore, given the broad-based support and government stimulus, we expect the following:

Hypothesis 1 (Green Benefits). *Stock prices increase after firms' green bond announcement.*

To understand the channels of and mechanism for the green bond issuance effect, we provide three hypotheses on the sources of the potential positive stock market reaction when firms announce their green bond issuance: the "investor attention" channel, the "fundamental" channel and the "financing cost" channel.

[15] See "green finance environmental impact is hard to measure" by Todd Cort and Cary Krosinsky, *Financial Times*, November 4, 2015.

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Investors have limited attention to corporate announcements (Ben-Rephael, Da and Israelsen (2017)), and a firm's visibility matters to stock market investors (Grullon, Kanatas, and Weston (2004)). When a firm announces its green bond issuance, media exposure will increase dramatically compared with conventional corporate bond issuance. Green bond issuers will issue formal press releases and declare that they have successfully labeled their newly issued bonds "green", which sends a good signal to the market.

Therefore, labeling green is a certificate or proof such that a firm's green projects and internal sustainable plans have been verified by a second party. In other words, the "label effect" will play a significant role when firms issue green bonds. Not only will bond investors discover this fact, stock investors will notice. More investor attention could potentially expand the investor base for issuers, which is good news for firms, and hence a positive announcement return.

If media attention plays a pivotal role inducing positive announcement return, then the market will only respond to first-time green bond issuance rather than every time a bond is labeled green. Green bond issuance will draw people's attention, attract more investors and we expect to see an increase in institutional ownership, which generates more trades on the stock market. Henceforth, stock liquidity improves after green bond issuance.

Hypothesis 2a (Investor Attention). *Green bond issuance attracts investors' attention and it will improve visibility and enlarge the investor base.*

An alternative explanation is the "fundamental channel", suggesting that green bond issuance contains more information about valuable investment opportunities (Myers and Majluf (1984), Kang and Stulz (1996)), which reduces information asymmetry and leads to the positive announcement effect. In particular, green bond issuers explicitly indicate their use of proceeds in the prospectus (see Appendix A) and show their ongoing or future environmental beneficial projects. Those statements will be examined externally by a second party to further address their green promise (see Appendix B). Under normal circumstances, when firms issue straight corporate bonds, they will not disclose information as much as they do when issuing green bonds. As a result, investors will benefit from additional information that the green bond issuer discloses when issuing a green bond so that the stock market will react positively to the announcement.

The fundamental mechanism suggests that every green bond issuance will contain valuable information and presumably the stock market will respond in every case. Meanwhile, because the

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investors believe in the firm's long-term high valuation, they will hold the stock rather than realize the gain in the short run so that stock liquidity may either shrink or hold steady. In sum, the liquidity test can differentiate "investor attention" channel from "fundamental" channel.

Hypothesis 2b (Fundamental). *Green bond issuance reveals valuable investment opportunities and illustrates the firm's dedication to sustainable development, which can help the issuing firm survive in the long run.*

Recently, researchers have shown that corporate social responsibility (CSR) exerts a significantly positive impact on firm value and financial performances both in the short run and in the long term. For example, among others, Sharfman and Fernando (2008), Ghoul, Guedhami, Kwok and Mishra (2011), Goss and Roberts (2011) and Chava (2014) show that CSR and better environmental, social and governance (ESG) performances are associated with a lower cost of loans, a lower cost of capital, an improved credit rating and superior financial performance.

Green bonds normally face oversubscription, and investors warm to those issuances.[16] From the stakeholder theory point of view, green bonds can be regarded as internalizing environmental externalities and catering to the appetite of a certain type of investor, especially investors with a green mandate. Thus, investors with a green mandate, who consider sustainable growth and environmentally friendly projects in addition to traditional risk and return, will appeal to the green bond market and push up the price. Investors have been analyzing green bonds because of increasing demand and recently, there has been some anecdotal evidence suggesting that new green bond issuers charge a premium because of that demand.[17]

Barber, Morse and Yasuda (2018) analyze 5000 VC funds and find that impact funds have a 14.1% higher probability of attracting investment, a phenomenon they attribute to growing market demand. Growing demand for green bonds indicates the lower cost of capital of issuing firms. Therefore, based on the financing cost mechanism, we expect green bonds to be priced at a premium (lower yield) in the primary market.

[16] https://www.wsj.com/articles/investors-warm-to-green-bonds-1491790201
[17] https://www.ft.com/content/9396fb28-ca2b-11e7-ab18-7a9fb7d6163e

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Hypothesis 2c (Financing Cost). *Green bond issuance attracts investors with a green mandate and socially responsible funds, which will push up the bond price and lower the cost of capital for the firm.*

3. Green Bond Market Data

3.1 Sample Construction

We construct the most comprehensive international green bond dataset augmenting the Climate Bond Initiative (CBI) with Bloomberg labeled green bonds. Among all the available data sources, these are the most inclusive sources and have a similar green bond selection criterion, including the use of proceeds, the selection of projects, eligibility verification, and reporting. We choose CBI because it has a larger coverage. For example, the first green bond widely accepted by the green bond market, the climate awareness bond issued by the European Investment Bank (EIB) in 2007, is covered in the CBI database but not in the Bloomberg green bond database. Nonetheless, there are several advantages of augmenting with Bloomberg, including an adequate amount of bond characteristics; furthermore, Bloomberg provides announcement dates, enabling us to conduct event study analysis.

Our CBI dataset cover the period from June 2007 to July 2017 and our Bloomberg data cover the period from March 2010 to December 2017. CBI defines green bonds as instruments created to fund projects that have positive environmental and/or climate benefits.[18] Proceeds from these bonds are earmarked for green projects but are backed by the issuers' entire balance sheet. They also include green "use of proceeds" revenue bonds, green project bonds, and green securitized bonds.[19] Due to data limitations, the CBI dataset only contains ISIN, issuer name, issue date, maturity date, amount issued, currency, bond type and domicile of the issuer. As of July 25, 2017, the total number of green bonds we obtained from CBI is 1181. The outstanding amount from their sources is USD 248 billion. The mean issue amount is approximately USD 211 million.

[18] The majority of green bonds in CBI are green "use of proceeds" or asset-linked bonds.
[19] https://www.climatebonds.net/market/explaining-green-bonds

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Approximately 42.8% of bond issuances are denominated in US dollars, and 14% of bonds are denominated in Euros.[20]

The second data source we use is Bloomberg, which includes four types of green bonds: the Green Use of Proceeds Bond, the Green Use of Proceeds Revenue Bond, the Green Project Bond, and the Green Securitized Bond. Bloomberg's definitions of green bonds are instruments for which the proceeds are *exclusively* applied (either by specifying use of proceeds, direct project exposure, or securitization) toward new and existing green projects, defined as projects and activities that promote climate or other environmental sustainability purposes. After cross-check and validation, we supplement the CBI data with Bloomberg bond information. As of December 29, 2017, the total number of bonds in the Bloomberg dataset is 1074. The mean issuance amount is USD 285 million and the average coupon is 3.26%. The merged dataset contains 1510 bonds worldwide during the 2007-2017 sample period.

We collect stock prices and market index prices from DataStream and firm financial data from Worldscope. We obtain matching sample data from WRDS Compustat Global financial and security prices. Institutional ownership is from FactSet Ownership ("Lionshares"), and we follow Ferreira, Miguel, and Pedro Matos (2008) to categorize different types of institutional investors. Green bond yield spread data and matched firm conventional bonds data are from the SDC Platinum New Issues database.

3.2 Descriptive Statistics

The first green bond is issued by the European Investment Bank on June 28, 2007 and the issue amount is EUR 600 million. The largest amount is a sovereign green bond issued by the Republic of France and the issue amount is EUR 7 billion.

[Insert Figure 1 here]

Figure 1 displays the green bond global market issuance over time. Whereas the first green bond was issued in 2007, the issue number and amount remain low in the first 5 years because development banks and supranational organizations are the only issuers of green bonds before

[20] We have also conducted a cross-validation with the Bloomberg data source to confirm the reliability of unique ISIN, issuer, and issue date. We dropped duplicates or other problematic bond entries from our final sample.

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2013. Green bond market experiences a skyrocketing boom in 2014 both in amount and in the number of issuances due to commercial banks and corporations. After 2015, the market grows steadily and continue to expand in 2016 and double in 2017. The CBI projects that the green bond market is expected to reach 250 billion annual issuances in 2018.

[Insert Figure 2 here]

Figure 2 shows that green bond issuers' geographical distribution, which is widely spread around the world and beneficial for green investors, especially institutional investors. According to Standard & Poor's estimation, green bonds are one of the best instruments to invest in green and sustainable projects.[21] Panel A of Table 1 shows that our initial sample includes 1510 unique observations of green bond issuance globally and includes 209 commercial banks, 456 corporations, 489 development banks, 41 state-backed entities, 209 municipal bonds, 4 sovereigns and 85 green asset-backed securities. Panel B shows that supranational organizations, including IFC, World Bank, EIB, and ADB, are among the most frequent issuers. China, US, and the European countries are the leading issue countries in the green bond market.

[Insert Table 1 here]

We only consider financials and industrial corporations, which accounted for 665 issuances. Then, we select those issuers that are public listed companies. Among those public issuers, we differentiate repeated issues from first-time issues because our media attention hypothesis suggests that when a firm announces its first green bond issuance, it will attract investors' attention. After first-time issuance, the media exposure effect will no longer exist because the firms have already been disclosed to the public. Finally, excluding subsequent issuance, our final sample is reduced to 132 unique public issuers.

[Insert Table 2 here]

Table 2 shows summary statistics for green bond characteristics. Among all green bonds, the average coupon is 3.3%, maturity is 7.6 years and average issue amount per bond is 286 million USD. The subsample of public issuers' green bonds, which are used for our regression analysis, however, have on average a shorter maturity but a larger issue amount.

[21] https://www.spglobal.com/our-insights/Beyond-Green-Bonds-Sustainable-Finance-Comes-of-Age.html

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4. Stock Price Reaction to Green Bond Announcement

4.1 Event Selection

We obtain the announcement date for green bond issuance from Bloomberg and/or reliable news reports (e.g., exchange announcements or official press releases); alternatively, the date is provided directly by the lead manager. For other green bonds whose announcements dates are not provided by Bloomberg, we manually search official company websites, press conferences, and news, whichever is earliest, to identify the closest announcement date. To alleviate the concern that bond issuance has some information linkage, we follow Kruger (2015) to conduct 10- and 21-day event windows. We then match the adjusted stock prices data from Datastream and company financial data from WorldScope. The final sample size is 132 unique public firms.[22]

To conduct our analysis, we exclude asset-backed green bonds and private placement. The reasons for this exclusion are that asset-backed green bond structures are different from normal green bonds, private placement green bonds' detailed financial information is not disclosed, and we cannot easily measure market reactions to a private placement. To obtain public issuers of green bonds, we first check whether the green bond issuer itself is a public listed company. If it is not, we consider whether the parent company of that bond issuer is a publicly listed firm. If the bond is issued in the same country in which the public stock is listed, then we include that stock. If not, we use the primary stock market in that firm's country. The findings are robust for different definitions of green bonds and sample selection. For example, when we use the CBI-only sample, our results are similar.

4.2 Event Study Analysis

To study stock reaction to bond issuance, we first use the CAPM model to estimate abnormal returns. Because our samples include the international green bond market, we must use data from different stock markets. We decide to use the index of the stock market on which the firm's stock is listed.

We first estimate the beta of each firm using estimation windows starting from 300 trading days to 50 trading days before the announcement date. Then, we use event windows from 10 days

[22] We list all the public issuers in our Internet Appendix IA.5.

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before and 10 days after the announcement. In addition, we run a robust check for event windows from 5 days before and 10 days after the announcement. We use index return minus treasury bond yield for market premium and stock return premium in the following regression, and abnormal returns for firm i and event day t are defined as follows:

$$R_{i,t} - R_{f,t} = a_i + \beta_i \times \left(R_{m,t} - R_f\right) + \epsilon_{i,t} \qquad (1)$$

$$AR_{it} = R_{it} - \hat{R}_{it} \qquad (2)$$

where a_i and β_i are all estimated parameters from the CAPM model. Column (1) of Panel A of Table 3 shows the result for the market reaction for the 21-day event window. We find that the stock market reacts positively and statistically significant (1.39%, *t*-statistic 2.36) around the green bond announcement. In Column (2), the stock market reacts from 5 days before to 10 days after the announcement date, and the CAR is approximately +1.04% (*t*-statistic is 2.12).

[Insert Table 3 here]

To further confirm stock market positively reacts to the green bond issuance, we conduct a nonparametric Wilcoxon signed-rank test. In Column (1), from 10 days before the announcement to 10 days after the announcement, the stock market's cumulative abnormal returns sign rank test P value is 0.0171, so we can reject the null hypothesis that CAR is not different from 0. Column (2) shows that we can reject the -5- to 10-day event window CAR also not different from 0 (P value 0.0303). The way to interpret the results is as follows: an event study on green bond issuance can be considered as one way to examine the information content of corporate news regarding environmental issues. The positive stock market reaction represents the "green label" effect, indicating that the firm's green projects are certified to have an environmental impact. Thus, the market will reward the company with a higher firm value. A similar pattern has been shown by Klassen and McLaughlin (1996), who demonstrated that firms receiving environmental awards will have a significant positive abnormal return afterward. From Figure 3, Google search volume peaked after Apple Inc. announced its green bond issuance on June 13, 2017, and search intensity remained high in the following week, which can also support our long event window arguments.

[Insert Figure 3 here]

In comparison, in the straight bond issuance announcement literature, Eckbo (1985) find that the stock market reaction to a straight bond issuance is associated with a slightly negative return

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in the shorter window. We are cautious about this argument because we consider a green bond announcement to involve two important aspects: the bond announcement effect and the "green label" effect. Normally, when researchers consider bond issuance, they typically analyze a short window around the event date. However, when researchers study an information disclosure or announcement of the positive/negative event, they tend to pay attention to a longer period. In this way, we present both the results and the longer period as more significant, so that we believe that the label effect is more important in green bond setting, which suggests that green bond issuance can be treated as an environmental award to green bond issuers. In addition, to visualize the positive stock market reaction, we show stock returns around the announcement date in Figure 4.

[Insert Figure 4 here]

For robustness, we use multi-factor models to calculate the abnormal return. We use Fama and French 3-factor (Fama and French, 1993) and 5-factor models (Fama and French, 2015). We find that in the 21-day event window, FF3 and FF5 are associated with 1.41% (t stat 2.42) and 1.29% (t stat 2.19), respectively. From 5 days before to 10 days after the announcement date, the stock market reacts to 1.06% and 1.01% in the FF3 and FF5 models. All the results are robust to the sign rank test.

In Columns (3) and (4) in Panel A of Table 3, we consider subsequent green bond issuance. According to investor attention, green bond issuers should not benefit from the label effect after their first-time green bond issuance. The reason for this is that after the company has successfully labeled a bond green and disclosed it to all investors, we should not find any positive effect for subsequent green bond issuance. On the contrary, the "fundamental" channel suggests investment opportunities when firms issue green bonds and it will react every time firms announce their issuances. As shown in Columns (3) and (4) of Panel B, the stock market reaction is not significant (0.12% and 0.61%, respectively), indicating that the "fundamental" channel seems not to be the driving force of positive announcement return.

Which types of green bond issuers are more likely to be rewarded by the stock market? To answer this question, we further differentiate the business sectors of green bond issuers. The reasons we conduct this analysis are twofold. First, the mechanisms between two major types of green bond issuers are different. Corporations issue green bonds to finance their own green projects, which is the core competency of their business. Another type of issuer is financial institutions, who

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will issue green bonds to make green loans to their borrowers or invest in other firms' green projects. There are fewer financial issues, but their amounts are larger. Second, we argue there is a need for studies involving CSR and ESG to further analyze the cross-industry effect. For instance, renewable energy companies might be much more exposed to environmental issues and benefit more than financial institutions.

We report the results for the financials and corporates test in Columns (1)-(4) in Panel B of Table 3. Columns (1) and (2) show the event study analysis for corporates. In Column (1), the CAPM model is 1.88% and statistically significant (t statistics 2.40). The results are similar for the FF3 and FF5 models. However, we find an insignificant stock market reaction for financials issuing green bonds. The results suggest that only firms conducting green projects will benefit from increasing valuation.

5. Channels and Mechanisms

In this section, we examine potential channels through which green bond issuance can affect the stock valuation of the issuers. We construct measures to examine those channels and then provide empirical evidence.

5.1 Green Premium

The "financing cost" channel implies that socially responsible investors will pursue holding green bonds to boost their ESG scores and thus to increase demand for those bonds. Correspondingly, green bond issuers can benefit from lower cost of capital and thus, the stock market reacts positively to the green bond announcement.

A basic measure of the green bond pricing benefit is yield spread at issuance. We download data from the SDC Platinum New Issues database. Our sample includes all the normal corporate bonds issued by the green bond issuers from 2007 to 2017, and we can conduct a within-firm comparison between green bonds and conventional bonds issued by the same firm. Moreover, we match a similar firm according to size, market to book and stock liquidity because higher liquidity tends to issue corporate bonds (Butler and Hong, 2010). Additionally, we must ensure that matched firms also have conventional bond issuances in the same year as green bond issuers. The choice of

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yield spread determinants is largely based on Campbell and Taksler (2003), Chen, Lesmond and Wei (2007) and Badoer and James (2016).

While there is anecdotal evidence suggesting a green bond premium,[23] the yield spread difference between and within firms is unclear. In Table 4, we regress the yield spread at the issuance date on the green bond dummy and control variables associated with bond characteristics and firm characteristics.

$$\begin{aligned} Yield\ \ Sprd_{i,t} = \beta_1 + \beta_1 \times Green_{it} + \beta_2 \times Equity\ \ Volatility_{it} + \beta_3 \times Bond\ \ Rating_{it} \\ + \beta_4 \times Maturiy_{it} + \beta_5 \times Issue\ \ Size_{it} + Firm\ \ Controls_{it} + Issuer\ \ FE \\ + Issuer\ \ Year\ \ FE + \epsilon_{it} \end{aligned} \quad (5)$$

where yield spread is the issue yield spread from SDC Platinum Global Issue database and green is an indicator equal to 1 if the bond is a green bond and 0 otherwise. Bond rating equals 1 if rating is lower than investment grade and 0 otherwise.

[Insert Table 4 here]

The fixed effects we consider include year by month, country, and issuer fixed effects. If we only add country fixed effects, we find a 6.9 bps premium, which is a very large pricing benefit for the firm, indicating that compared with other conventional bonds issued by the match firms, green bonds have pricing benefits. However, when we add firm fixed effect and year by month fixed effects, which only examine bond yield spread differences issued by the same issuer within the same year and month, the yield spread is no longer statistically significant. Therefore, the "financing cost" channel seems not to be the driving force of positive announcement returns. We note that only 41 firms have issued both regular corporate bonds and green bonds during our sample period. Such a relatively smaller sample may limit the power of the test.

5.2 Institutional Ownership

The "investor attention" channel suggests increasing media exposure when firms announce their green bond issuance. This exposure will send a good signal to the market about such firms' green investment opportunities therefore enlarge their investor base. As shown in Figure 5, there is higher stock market turnover around the green bond issuance month.

[23] https://www.ft.com/content/9396fb28-ca2b-11e7-ab18-7a9fb7d6163e

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[Insert Figure 5 here]

Institutional ownership data are from the FactSet (LionShares) database for the period from 2007 to 2017. Institutional investors contained in the database can be classified into different sub-categories, including mutual funds, pension funds, hedge funds, banks and insurance companies. For equities traded in the United States, the stock holding data are from 13F mandatory filings required by SEC. For equities traded in countries other than the US, FactSet gathers all the information from company announcements, regulatory agencies and annual reports. We also use a matching sample and conduct a diff-in-diff analysis, and the criterion is the same as in the previous section.

Our empirical test regarding institutional ownership includes both a green dummy, a post dummy and their interaction to analyze the institutional ownership change after green bond issuance compared with conventional corporate bond issuance. We follow Ferreira and Matos (2008) and Chung and Zhang (2011) in our empirical design. We include country and year fixed effects in all the specifications in Table 5. Our specification is as follows:

$$
\begin{aligned}
& Inst\ \ own_{i,t} \\
& = \beta_0 + \beta_1 \times Green_{it} \times Post_{it} + \beta_2 \times Green_{it} + \beta_3 \times Post_{it} + \beta_4 \times Size_{it} \\
& + \beta_5 \times Leverage_{it} + \beta_6 \times Profitability_{it} + \beta_7 \times Cash_{it} + \beta_8 \times R\&D_{it} + \beta_9 \times Sale_{it} \\
& + \beta_{10} \times Dividend_{it} + Country\ \ FE + Year\ \ FE \\
& + \epsilon_{it} \qquad (6)
\end{aligned}
$$

In Table 5, all the specifications are positively significant. In Column 2, after green bond issuance, institutional ownership increases by 7.9% compared with conventional bonds issued by the match firms.

[Insert Table 5 here]

Moreover, there is a home bias for domestic investors (Coval and Moskowitz (1999), Schumacher (2018)) and they are more subject to the attention-driven effect than foreign investors. In Table 6, according to Ferreira, Miguel, and Pedro Matos (2008), we separate domestic institutional investors from foreign institutional investors. Columns 1 and 2 of Table 6 suggest that only domestic investors and domestic independent investors increase their holdings in firms that

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issue green bonds by 8.5% (robust t-statistics 3.20) and 7.6%, respectively, which is both statistically and economically significant.

[Insert Table 6 here]

5.3 Stock Liquidity

The "firm fundamental" channel implies that green bonds illustrate the firm's valuable investment opportunities. Furthermore, green projects associated with future positive cash flow can help firms survive in the long run. We adopt stock liquidity tests to differentiate the "fundamental" channel from the "investor attention" channel.

We use the quoted bid-ask spread divided by the mid-point of the bid-ask price as a proxy for stock liquidity. In addition to the widely used quoted spread measure, we follow Amihud's (2002) methods to calculate stock liquidity and use the absolute return scaled by dollar trading volume and average among different periods to represent the illiquidity of stock. Fong, Holden, and Trzcinka (2017) analyze the best liquidity proxy for global research and determine that the Amihud measure provides the best monthly cost-per-dollar-volume proxies. The detailed calculation method is as follows:

$$Illiquidity_i = \frac{1}{Days_i}\sum_{t=1}^{Days_i}\frac{|Return_{it}|}{Volume_{it}} \qquad (7)$$

The Amihud illiquidity measure is a daily high-frequency measure, so we will take the arithmetic mean of all the trading days' daily illiquidity to obtain the monthly Amihud illiquidity. Through a simple univariate analysis, we find that liquidity increases after issuing a green bond. In our full sample analysis, both of our measures of liquidity increase after issuing green bonds.[24] Next, we construct a balanced sample for one year before and one year after issuing green bonds. We an un-tabulated univariate test for which the results are available upon request.

5.3.1 Panel Regressions

Now we conduct a panel regression and control for firm characteristics, liquidity control variables and firm and year fixed effects. The specification we use for a panel regression is as follows:

$$Liquidity_{it} = a + b \times Post_{it} + FirmCharacteristics_{it} + FE + \epsilon_{it} \qquad (8)$$

[24] This means that both the quoted spread and the Amihud illiquidity measure decrease.

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Illiquidity is a log transferred measure: the larger the value, the less liquid the stock. The post dummy equals zero before bond issuance and one otherwise. FE includes firm fixed and year fixed effects. The results are shown in Table 7.

[Insert Table 7 here]

We use the Amihud method as a robustness check because Lesmond (2005) suggests that the Amihud method performs well in cross-country analysis. The reason we take the logarithm of Amihud is because the liquidity measures are highly skewed. Although this approach does not change our linear estimator, it will bias our statistical inference so that we take the logarithm to obtain a normal distribution of our dependent variable. We calculate the daily Amihud measure and take the average in that calendar month. After issuing green bonds, green bond issuers' stock liquidity improves significantly in the one-year horizon. Not only is this statistically significant, we can recognize that the result is economically meaningful. In the baseline result in Column 1 of Table 7, one year after green bond issuance, issuers' stock liquidity improves by 4.87% (exp(-0.0546)-1) compared with before. Next, we add firm fixed and year fixed effects to control for unobserved factors. Additionally, we calculate the heteroskedasticity-consistent robust standard error, and the results are consistent.

In Column 2 of Table 7, we add firms' characteristics as control variables and find out that after issuance, quoted spread liquidity measure decrease -8.3%, we take the exponential of that and interpret it as stock liquidity increases by 7.96%. In Column 4 of Table 7, we find that stock market liquidity improves 11.9% (exp(-0.127)-1) after issuance.

5.3.2 Matching Sample and Diff-in-Diff Analysis

The results have already been addressed with some concerns about endogeneity; however, the liquidity trend could still bias our results if all firms' liquidity has improved in the green bond issue year. First, we argue that this is an international study, and we do not expect to see the same event occur at the same time and have a positive impact on stock liquidity. Second, to further alleviate this concern, we analyze a matching sample and test conditionally on whether liquidity improves and whether green bond issuers would benefit more than conventional bond issuers. We use a matching method and choose criteria according to their size, market-to-book, and previous year liquidity. We restrict the matching firm sample listed on the same exchange as green bond issuers.

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We impose further restrictions on matching firms' criteria. There is no evidence to suggest that after corporate bond issuance, stock liquidity will increase. However, there could be some omitted variables that induce the firms with a higher propensity to issue bonds while leading to future stock liquidity increases. To tackle those issues, we not only match the sample according to size, market-to-book, previous year liquidity and in the same industry, same country, same exchange but also restrict the matching firm sample with bond issuances the same year.

[Insert Figure 6 here]

In Figure 6, we show that prior to green bond issuance, green bond issuers and match firms' stock liquidity trend are parallel and have no significant differences. Nevertheless, after green bond issuance, green bond issuers' liquidity increases much more than straight corporate bonds issued by the match firms. We further run a diff-in-diff analysis to confirm our hypothesis, and our regression model is as follows:

$$Liquidity_{it} = a + b * Post_{it} + c * Green_{it} + d * Post_{it} * Green_{it} + Firm\ \ Controls_{it} + Year\ \ FE + \epsilon_{it}$$

Post is a dummy and equals one after bond issuance; before issuance it is zero. Green is also a dummy that equals one for green bond issuers and zero for non-green bond issuers. We find that post-issuance liquidity is improving and green bond issuers' stock liquidity increases more than 1.27% compared with match firms, as shown in Column 2 of Table 8. In the Amihud measure, we find a 2.2% stock liquidity increase for green bond issuers compared with match firms. To further confirm our hypothesis, we test on a shorter horizon for diff-in-diff methods. We study 3 months and 6 months before and after the announcement date, and the results are robust.[25]

[Insert Table 8 here]

5.3.3 Conditioning on Initial ESG Profiles

If the media attention channel leads to a positive abnormal return and subsequent stock liquidity improvement, then the firms' prior ESG profiles will have different effects. In particular, companies that have been already been exposed to ESG attention, that are under examination by a third party, or that have relatively high ESG profiles will benefit less from green bond issuance.

[25] See Internet Appendix IA.4 for the estimation results.

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Their ESG profiles have been understood by investors such that they will not attract as much attention as firms with less ESG coverage. To confirm this hypothesis, we interact ESG coverage with Green*Post term and analyze heterogeneous effects on prior ESG profiles status. The specification is as follows:

$$Liquidity_{it} = a + b * Post_{it} * Green_{it} * ESG\ \ Coverage_{it} + b * Post_{it} * Green_{it} + c * Post_{it} + d * Green_{it} + e * ESG\ \ Coverage_{it} + Firm\ \ Controls + Country\ \ FE + Year\ \ FE + \epsilon_{it}$$

ESG coverage equals 1 if the firm is covered by RepRisk ESG database at time t. In Columns (2) and (4) of Table 9, we include controls and fixed effects, and standard errors are clustered at the firm level. In Column (1), green bond issuance will increase the firm's stock liquidity by 1.95%; however, if the firm has already been covered by the ESG database, the liquidity improvement will be 0.34% less (*t*-statistic=2.849). Additionally, the results are robust to the Amihud illiquidity measure.

[Insert Table 9 here]

6. Conclusion

In this paper, we construct a comprehensive dataset covering all corporate green bond issuance worldwide to provide the first empirical analysis of the market's reaction to firms' environmental, social, and governance (ESG) activities. Green bond issuance can be viewed as a proxy for firms to make environmentally friendly investments and change their ESG profiles. We find that the issuers' stock prices increase significantly around the announcement of green bond issuance. Stock market reactions are stronger for first-time issuers than for repeated issuers and stronger for corporate issuers than for financial institution issuers.

We explore three potential sources of positive announcement returns. We find little evidence that green bonds are issued at lower yields than regular corporate bonds from the same issuers, suggesting that the main advantage of green bonds is not cheaper debt financing. However, we find increased institutional ownership and improved stock liquidity after green bond issuance. Green bonds can help enlarge the investor base because issuing green bonds can attract more media exposure and be used by impact investors to satisfy their investment mandates. Overall, our findings suggest that existing shareholders derive net benefits from green bond issuance.

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Appendix A: Excerpt of Apple Green Bond Prospectus

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Prospectus Supplement
(To Prospectus dated April 29, 2013)

$12,000,000,000

Apple Inc.

$500,000,000 Floating Rate Notes due 2019
$500,000,000 Floating Rate Notes due 2021
$500,000,000 1.300% Notes due 2018
$1,000,000,000 1.700% Notes due 2019
$2,250,000,000 2.250% Notes due 2021
$1,500,000,000 2.850% Notes due 2023
$2,000,000,000 3.250% Notes due 2026
$1,250,000,000 4.500% Notes due 2036
$2,500,000,000 4.650% Notes due 2046

We are offering $500,000,000 of our Floating Rate Notes due 2019 (the "2019 Floating Rate Notes"), $500,000,000 of our Floating Rate Notes due 2021 (the "2021 Floating Rate Notes" and, together with the 2019 Floating Rate Notes, the "floating rate notes"), $500,000,000 of our 1.300% Notes due 2018 (the "2018 Fixed Rate Notes"), $1,000,000,000 of our 1.700% Notes due 2019 (the "2019 Fixed Rate Notes"), $2,250,000,000 of our 2.250% Notes due 2021 (the "2021 Fixed Rate Notes"), $1,500,000,000 of our 2.850% Notes due 2023 (the "2023 Fixed Rate Notes"), $2,000,000,000 of our 3.250% Notes due 2026 (the "2026 Fixed Rate Notes"), $1,250,000,000 of our 4.500% Notes due 2036 (the "2036 Fixed Rate Notes") and $2,500,000,000 of our 4.650% Notes due 2046 (the "2046 Fixed Rate Notes" and, together with the 2018 Fixed Rate Notes, the 2019 Fixed Rate Notes, the 2021 Fixed Rate Notes, the 2023 Fixed Rate Notes, the 2026 Fixed Rate Notes and the 2036 Fixed Rate Notes, the "fixed rate notes"). We refer to the floating rate notes and the fixed rate notes collectively as the "notes."

The 2019 Floating Rate Notes will bear interest at a floating rate equal to three-month LIBOR plus 0.820% and mature on February 22, 2019. The 2021 Floating Rate Notes will bear interest at a floating rate equal to three-month LIBOR plus 1.130% and mature on February 23, 2021. We will pay interest on the floating rate notes quarterly in arrears on February 23, May 23, August 23 and November 23 of each year, beginning on May 23, 2016.

We will pay interest on the fixed rate notes semi-annually in arrears on February 23 and August 23 of each year, beginning on August 23, 2016. The 2018 Fixed Rate Notes will mature on February 23, 2018, the 2019 Fixed Rate Notes will mature on February 22, 2019, the 2021 Fixed Rate Notes will mature on February 23, 2021, the 2023 Fixed Rate Notes will mature on February 23, 2023, the 2026 Fixed Rate Notes will mature on February 23, 2026, the 2036 Fixed Rate Notes will mature on February 23, 2036 and the 2046 Fixed Rate Notes will mature on February 23, 2046.

We may redeem the fixed rate notes in whole or in part at any time or from time to time at the redemption prices described under the heading "Description of the Notes—Optional Redemption" in this prospectus supplement. The floating rate notes may not be redeemed before maturity. The notes will be issued only in minimum denominations of $2,000 and integral multiples of $1,000 in excess thereof.

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USE OF PROCEEDS

Use of Proceeds for Floating Rate Notes and 2018, 2019, 2021, 2026, 2036, and 2046 Fixed Rate Notes

We estimate the net proceeds from sales of the notes (other than the 2023 Fixed Rate Notes) will be approximately $10.44 billion, after deducting underwriting discounts and the *pro rata* offering expenses for such notes. We intend to use such net proceeds for general corporate purposes, including repurchases of our common stock and payment of dividends under our program to return capital to shareholders, funding for working capital, capital expenditures, acquisitions and repayment of debt.

We may temporarily invest funds that are not immediately needed for these purposes in short-term investments, including cash, cash equivalents and/or marketable securities.

Use of Proceeds for 2023 Fixed Rate Notes

We estimate the net proceeds from sales of the 2023 Fixed Rate Notes will be approximately $1.49 billion, after deducting underwriting discounts and the *pro rata* offering expenses for the 2023 Fixed Rate Notes. An amount equal to such net proceeds will be allocated for investments in one or more Eligible Projects (as defined below), including new and ongoing development of such Eligible Projects. Eligible Projects may include projects of our subsidiaries that meet the Eligibility Criteria set forth below.

We plan to identify Eligible Projects that fall within three environmental priorities where we believe we can make the most environmentally positive impact:

1. Reduce our impact on climate change by using renewable energy sources and driving energy efficiency in our facilities, products and supply chain.
2. Pioneer the use of greener materials in our products and processes.
3. Conserve precious resources.

"*Eligible Projects*" means projects that meet the Eligibility Criteria (as defined below).

"*Eligibility Criteria*" means any of the following:

- expenditures related to new and ongoing renewable energy projects, such as solar and wind projects, or associated energy storage solutions,
- expenditures related to projects that have received within the last three years, or are expected to receive, certification of LEED Gold or Platinum or BREEAM Very Good, Excellent, or Outstanding "green building" standards,
- expenditures related to the implementation of environmental design elements for new or ongoing building developments, such as high performance mechanical systems, natural ventilation, on-site renewable energy, and high performance lighting systems,
- expenditures related to energy efficiency projects and technologies at our corporate facilities, such as heating, ventilation and air

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Appendix B: Excerpt of Third-party Opinion on Apple Green Bond by Sustainalytics

APPLE INC. GREEN BOND

FRAMEWORK OVERVIEW AND SECOND-PARTY OPINION BY SUSTAINALYTICS

February 16, 2016

© Sustainalytics 2016

based materials, and/or recyclable materials and are not considered toxic substances (according to its Regulated Substance Specification).

Eligible Projects

Any project that meets the above eligibility criteria, including the new and ongoing developments of such projects, are eligible to be funded in whole or in part by an allocation of the green bond proceeds.

Eligible projects may include projects of Apple's subsidiaries that meet the eligibility criteria.

2.2.1 Project Selection Process

Apple's Environment, Policy and Social Initiatives (EPSI) team will:

(i) assess and determine project eligibility; and,
(ii) recommend an allocation of proceeds to eligible projects.

These projects are developed and managed by central teams, which are primarily based in Cupertino. For example, Apple has a Global Energy Team that manages all corporate renewable energy and energy efficiency projects; centralized Real Estate & Facilities teams that track and plan for all facilities worldwide, including green buildings and its new campus; and a Global Data Centre team that manages the global expansion of its data centres.

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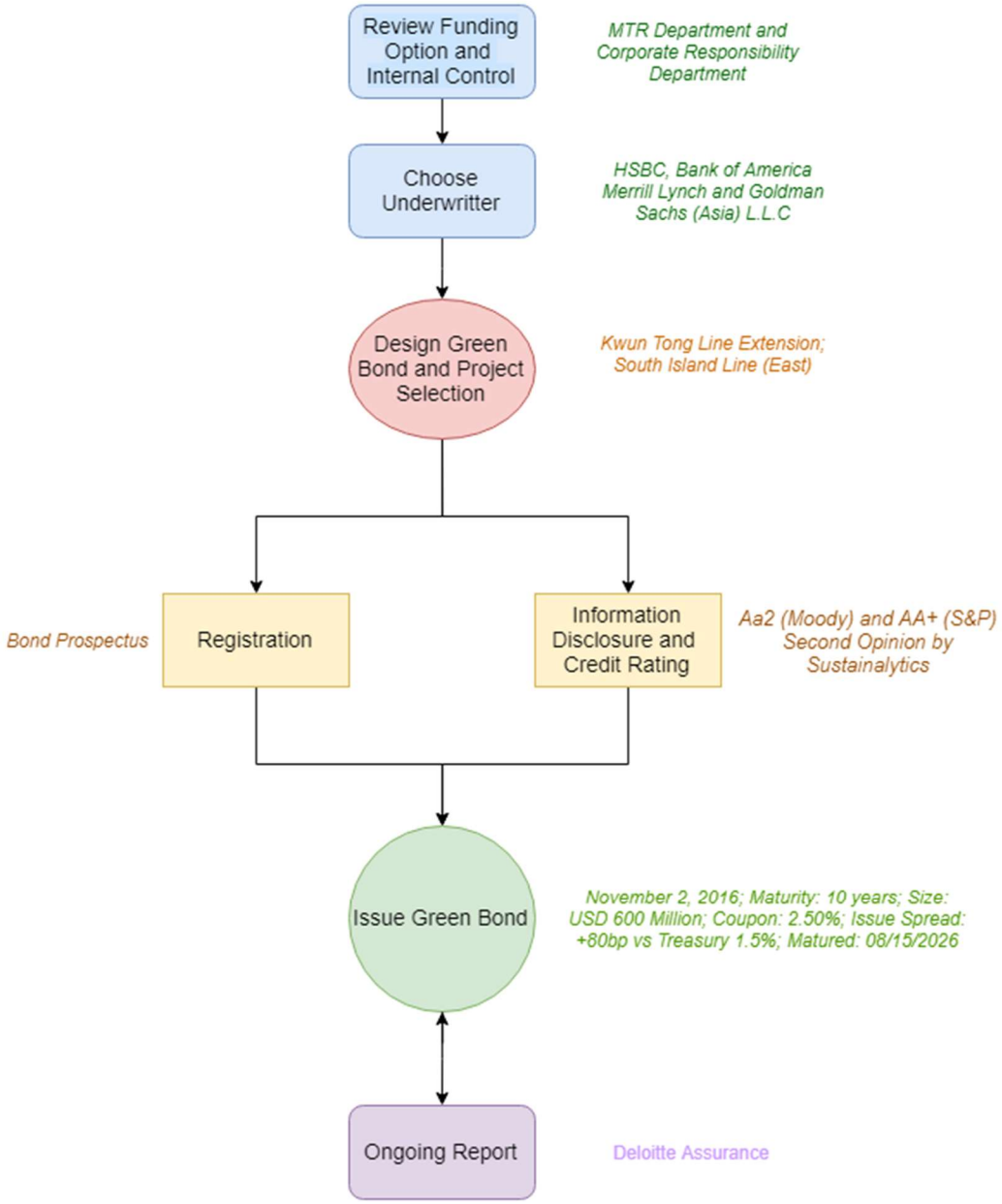
Appendix C: Green Bond Issuance Process: Hong Kong MTR



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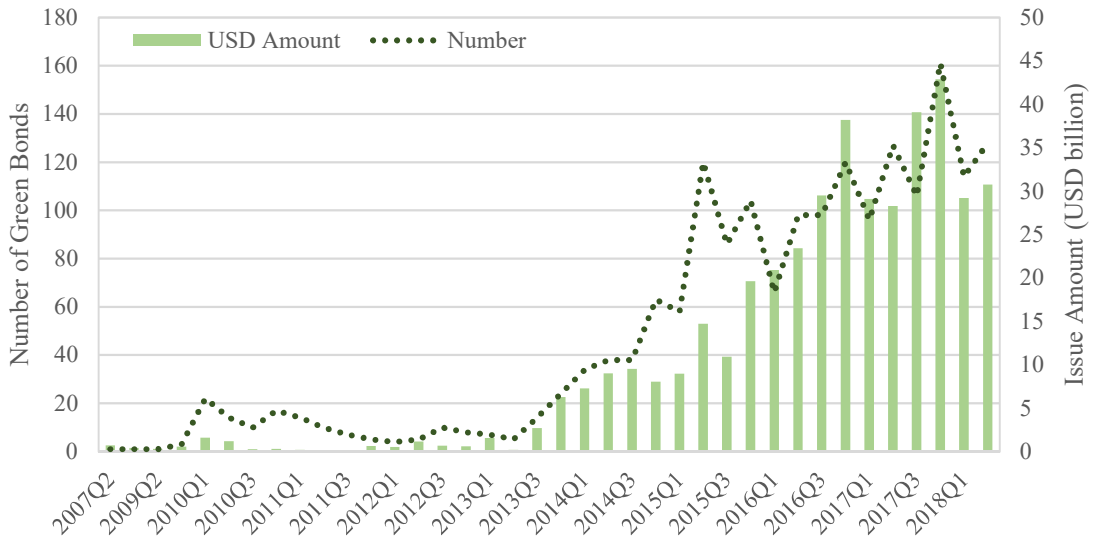

Figure 1. Global green bond issuance over time.
This figure shows the number and USD amount of green bond issuance on a quarterly basis from 2007Q2 (issuance of the first green bond by EIB) to 2018 Q2. We merge the Climate Bond Initiative (CBI) database with the Bloomberg green bond dataset to construct an international green bond dataset.

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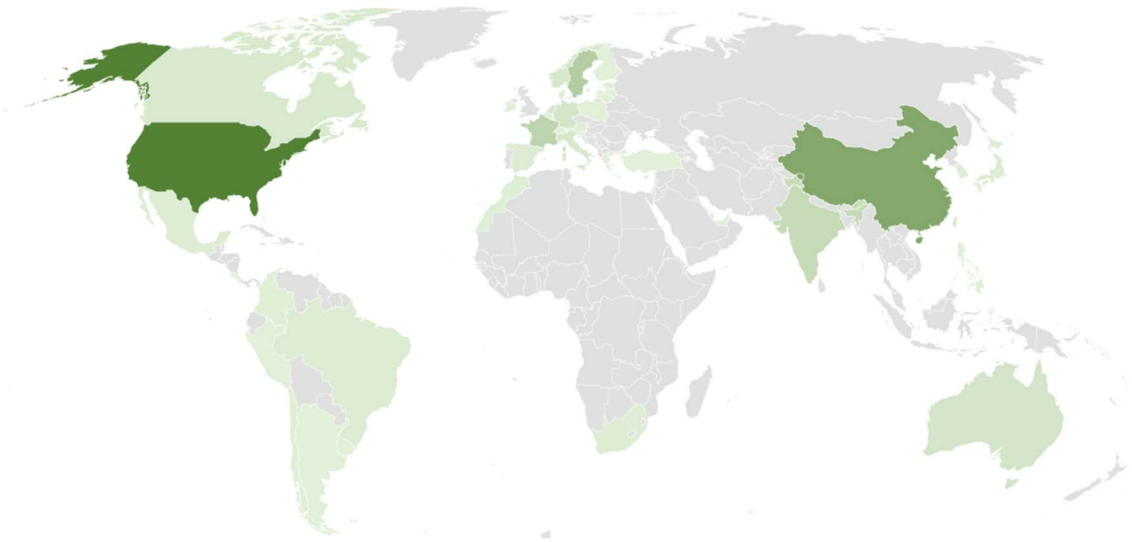

Figure 2. Geographical distribution of green bonds.
This figure shows that countries around the globe have been issued green bonds until 2018 Q2. We merge the Climate Bond Initiative (CBI) database with the Bloomberg green bond dataset to construct an international green bond dataset. The darker the green, the more issuances of green bonds in a country.

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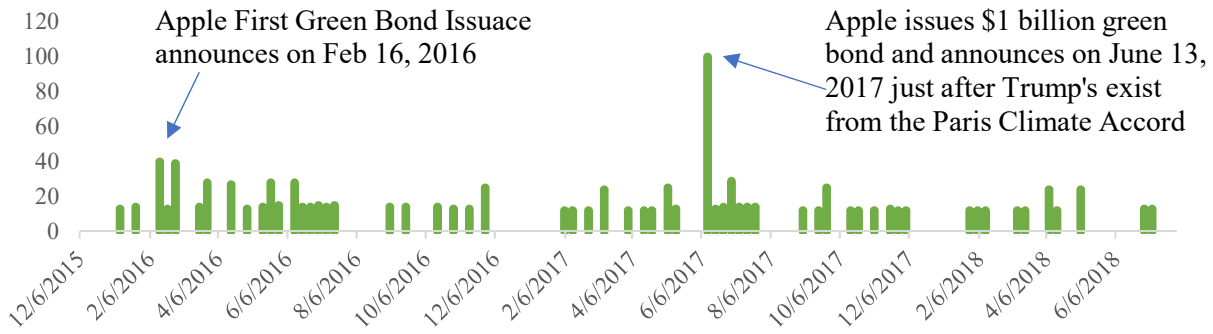
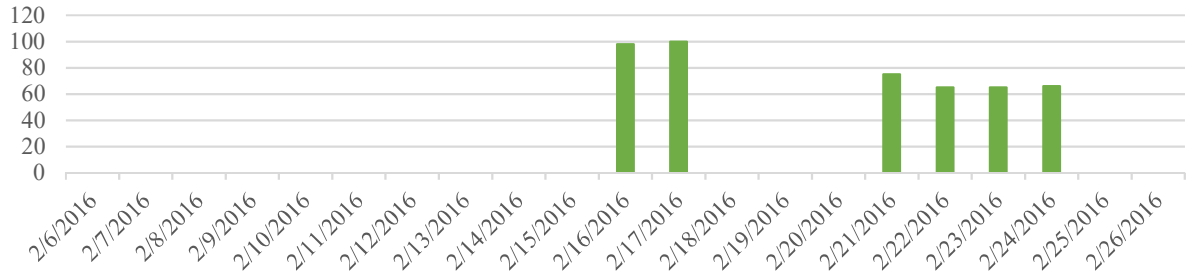
Panel A. "Apple green bond" Google search volume



Panel B. Apple first green bond issuance Google search volume zoom in



Panel C. Apple issues green bond after Trump's Paris Climate Accord exit zoom in

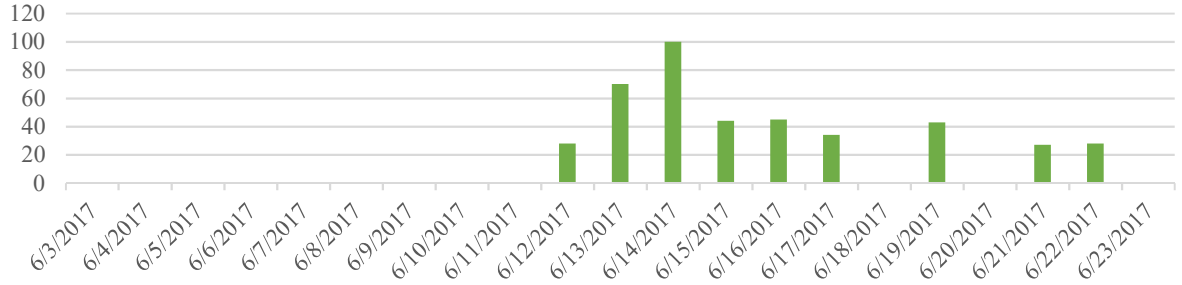


Figure 3. Apple green bond Google search volume.
This figure shows search intensity of "Apple Green Bond" on Google. Time period is from Jan 1, 2016, to August 1, 2018.

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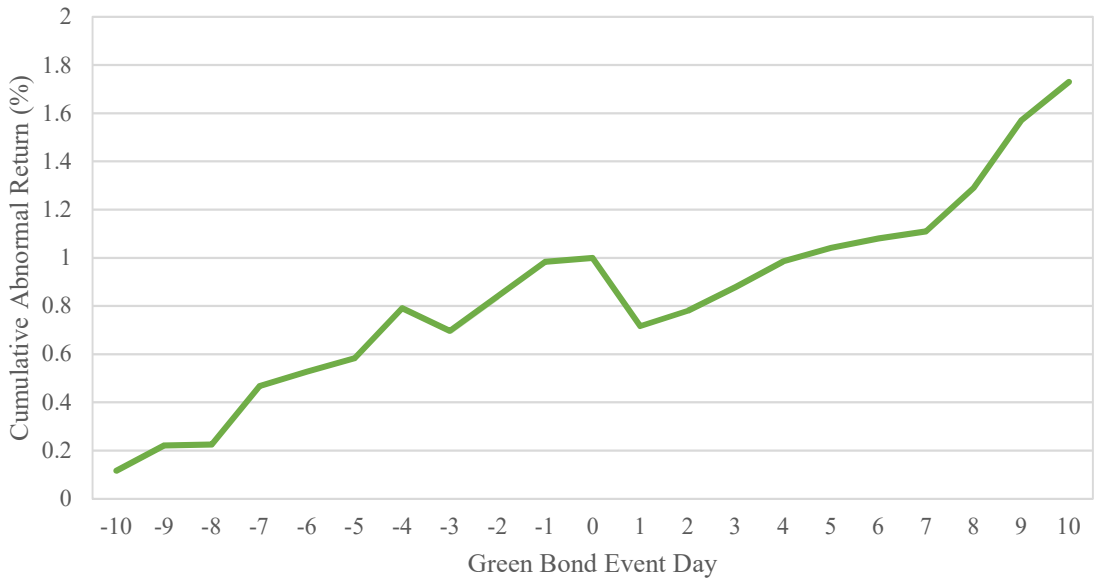

Figure 4. Stock market reaction to green bond issuance announcement.
This figure illustrates the stock market reaction to the green bond announcement around a 20-day event window. We use CAPM to calculate abnormal return, and the estimation window is from [-300, -50].

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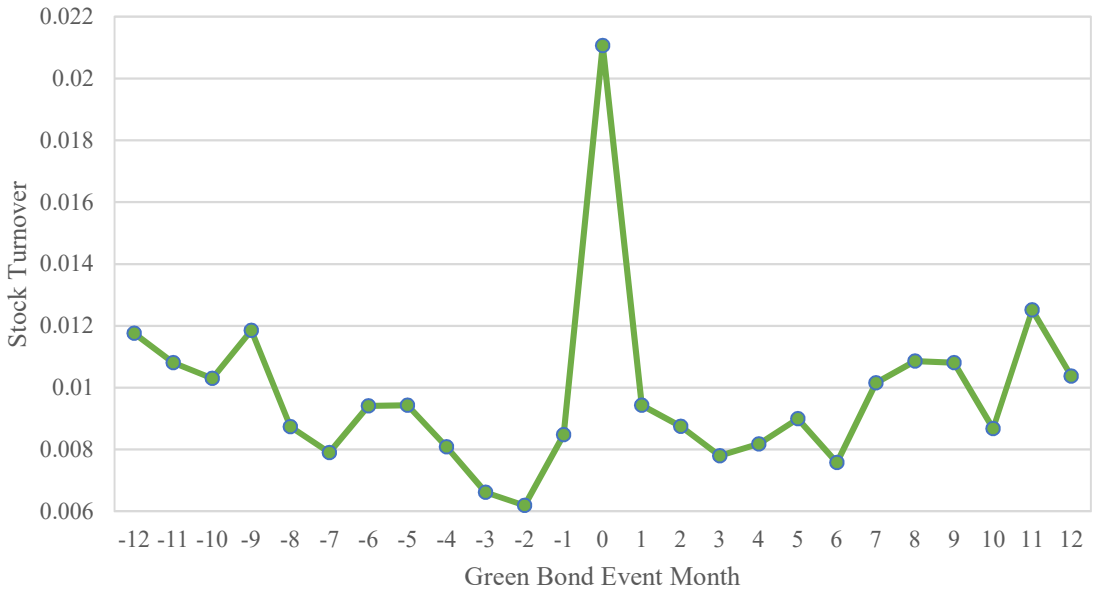

Figure 5. Average daily turnover around issuance of green bonds.
This figure shows green bond public issuers' trading behavior. Vertical axis is average turnover by volume (VO from Datastream) in a given month scaled by total shares outstanding. Time period is from Jan 1, 2016, to August 1, 2018.

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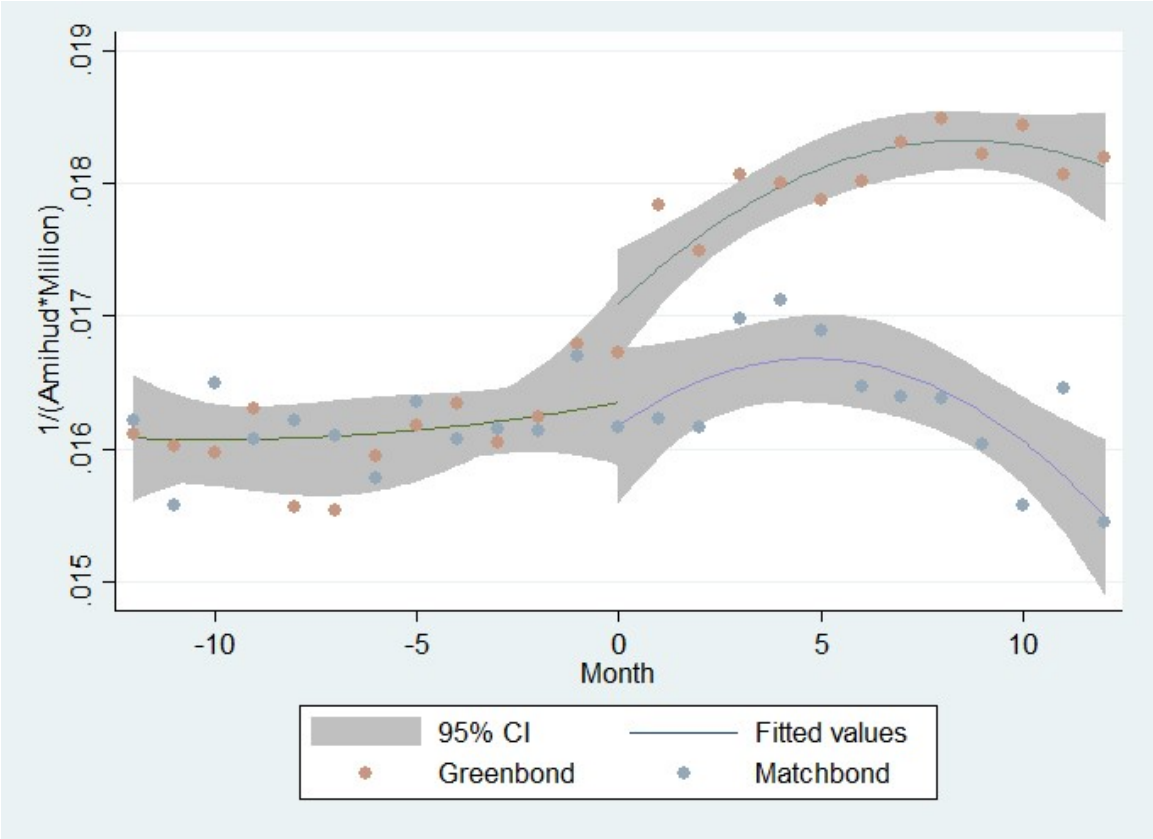

Figure 6. Stock liquidity before and after green bond issuance.
This figure shows stock liquidity in terms of the Amihud measure (absolute return over trading volume) before and after the green bond issuance compared with matched firms. We match firms according to size, book to market, industry, and previous year stock liquidity. More importantly, we select matching firms that also issue normal corporate bonds in the same year as green bond issuers to alleviate the concern that bond issuance generally will improve stock liquidity.

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Table 1. Green Bond Sample

Panel A shows the issuers' type decomposition of green bonds. The first green bond issuer in the world is the European Investment bank, which is classified as a development bank in our sample. The bonds are issued at a face value of €100 each and pay no coupons during the term. The bonds have full capital protection. At maturity, the bonds have full redemption of the face value plus a minimum payment of 5% − i.e., a minimum redemption of €105 per bond. The first three corporate green bonds are issued in November 13 by Bank of America, Vasakronan, and EDF, a French electricity-generating giant. In this paper, we focus only on public listed firms. Panel B shows the final sample construction and illustrates that we finalize our sample to 132 unique public issuers.

Panel A: Green Bond by Issuer Type and by Year

	Development bank	Municipal	Commercial bank	Corporation	State backed	ABS	Sovereign	All
2007	1	0	0	0	0	0	0	1
2008	1	0	0	0	0	0	0	1
2009	4	0	0	0	0	0	0	4
2010	57	5	0	0	0	0	0	62
2011	33	3	0	0	0	0	0	36
2012	23	3	2	0	0	0	0	28
2013	29	4	11	3	2	1	0	50
2014	65	24	28	43	3	7	0	170
2015	105	46	21	178	2	13	0	365
2016	80	59	62	103	9	41	1	355
2017	91	82	85	129	25	23	3	438
Total	489	226	209	456	41	85	4	1510

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Table 1. Green Bond Sample — *Continued*

Panel B: Green Bond by Country

COUNTRY	Full Sample	# of bonds issued by public issuers	Unique public issuers
SUPRANATIONAL	261	N.A.	N.A.
CHINA	242	54	32
UNITED STATES	195	21	17
SWEDEN	163	32	8
FRANCE	121	28	9
MALAYSIA	54	0	0
GERMANY	64	6	2
NETHERLANDS	43	14	3
NORWAY	35	8	2
CANADA	18	6	3
INDIA	17	9	8
AUSTRALIA	15	7	5
JAPAN	15	9	6
SPAIN	14	7	2
TAIWAN	14	5	5
BRITAIN	13	5	4
SINGAPORE	12	2	2
SOUTH KOREA	12	1	1
HONG KONG	11	5	3
MEXICO	11	1	1
Others	180	21	19
Total	1510	241	132

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Table 2. Summary Statistics

This table reports summary statistics for green bonds, public issuers and liquidity measures key variables. Public issuer characteristics use the three-year average of all the accounting data from Compustat Global. The sample consists of all unique firms and the sample period is from 2007Q2 to 2017Q4.

Panel A. Bond Characteristics

	Mean	Median	Std.	N
		All Green Bonds		
Coupon (Percent)	3.3	2.8	2.4	1510
Maturity (Year)	7.62	5	6.45	1510
Amount (million)	286	75	575	1510
		Public Issuers' Green Bonds		
Coupon (Percent)	3.2	2.9	2.2	241
Maturity (Year)	6.56	5	4.21	241
Amount (million)	369	153	552	241

Panel B. Firm Characteristics

Variable names	Mean	Median	Std.
Leverage	0.375	0.335	0.214
Total Asset(ln)	16.93	16.65	2.482
Profitability	0.010	0.014	0.185
Tangibility	0.329	0.212	0.342
Book Leverage	0.375	0.335	0.214
Market Leverage	0.315	0.288	0.186
Market to Book	1.250	1.035	0.589
Cash to asset	0.101	0.073	0.099

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Table 3. Stock Market Reaction to Green Bond Issuance Announcement

This table reports the CAR for green bond announcement return. We calculate the cumulative abnormal return (CAR) using the CAPM model. The stock price and market index are from Datastream and Bloomberg. We use the risk-free rate as the ten-year treasury bond yield provided by Bloomberg. We use a 250-day estimation window starting from 300 days before the announcement to 50 days before the announcement. We test for cross-sectional CAR significantly different from zero. P (Sign-test) is the P value for the sign-test and it is non-parametric. (Robust t statistics in parentheses*** p<0.01, ** p<0.05, * p<0.1.)

Panel A: First-time vs. Subsequent issues

	First-time issue		Subsequent issues	
	(1)	(2)	(3)	(4)
Event Window	[-10,10]	[-5,10]	[-10,10]	[-5,10]
CAPM_CAR (%)	1.39**	1.04**	0.12	0.61
	(2.36)	(2.12)	(0.21)	(0.99)
P(Sign-test)	0.0171**	0.0303**	0.5661	0.8667
FF3_CAR (%)	1.41**	1.06**	0.10	0.31
	(2.42)	(2.24)	(0.14)	(0.60)
P(Sign-test)	0.0247**	0.0225**	0.7723	0.7846
FF5_CAR (%)	1.29**	1.01**	0.50	0.23
	(2.19)	(2.07)	(0.76)	(0.40)
P(Sign-test)	0.0513*	0.0492**	0.3014	0.7253
Observations	132	132	109	109

Panel B: Corporates vs. Financials

	Corporate		Financial	
	(1)	(2)	(3)	(4)
Event Window	[-10,10]	[-5,10]	[-10,10]	[-5,10]
CAPM_CAR (%)	1.88**	1.51**	0.44	0.15
	(2.40)	(2.22)	(0.55)	(0.26)
P(Sign-test)	0.0132**	0.0264**	0.5962	0.5962
FF3_CAR (%)	1.96**	1.59**	0.36	0.09
	(2.54)	(2.42)	(0.45)	(0.15)
P(Sign-test)	0.0160**	0.0152**	0.6820	0.6114
FF5_CAR (%)	1.86**	1.53**	0.22	0.04
	(2.36)	(2.25)	(0.27)	(0.08)
P(Sign-test)	0.0302*	0.0342*	0.8741	0.8058
Observations	86	86	46	46

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Table 4. Green Bond Yield Spread

The table presents a yield spread analysis of green bond issuance. The dependent variable is yield spread in percentage at issuance date. The control pool includes matched control firms' issuance of normal green bonds and the same firms' issuance of normal corporate bonds. We collect all the bonds from SDC Platinum Global Issuance database. The sample starts in 2007 and ends in 2017. Size is the log of market capitalization of the firm. Leverage refers to the leverage ratio, calculated as the firm's total debt divided by the market value of total assets. Cash is the ratio of cash and short-term investments to total assets. Heteroscedasticity-consistent *t*-statistics clustered at the firm level are reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	Yield Spread			
	(1)	(2)	(3)	(4)
Green	-0.0694**	-0.0639*	-0.0588	-0.008
	(-2.19)	(-1.81)	(-1.62)	(-0.02)
Equity Volatility	45.365***	47.383***	19.707*	3.144
	(5.76)	(5.90)	(1.95)	(0.16)
Bond rating (BBB or worse)	0.274**	0.252**	0.058	-0.198
	(2.57)	(2.30)	(0.49)	(-1.63)
Maturity	0.057***	0.059***	0.059***	0.054***
	(6.97)	(7.14)	(6.59)	(5.98)
Issue Size	-0.256***	-0.240***	-0.221***	-0.205***
	(-11.06)	(-9.67)	(-8.21)	(-7.67)
Firm Size			-2.976***	-0.350
			(-7.73)	(-0.69)
Leverage			0.772***	-0.095
			(4.98)	(-0.47)
Profitability			0.150	-0.148
			(1.27)	(-0.98)
Tangibility			0.175	0.189
			(1.58)	(1.52)
Country FE	YES	NO	NO	NO
Issuer FE	NO	YES	YES	YES
Year by Month FE	NO	NO	NO	YES
No. firms issue both bonds	41	41	41	41
R-squared	0.311	0.356	0.278	0.386
Observations	2435	2435	2435	2435

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Table 5. Institutional ownership diff-in-diff analysis

This table presents the results of institutional investors' ownership change after firms have successfully labeled their bonds green. Following Ferreira, Miguel, and Pedro Matos, 2008, the data range from 2000 to 2017. Institutional ownership (IO) is defined as total institutional ownership in percentage of market capitalization. Institutional ownership dollar value is defined as log of total institutional ownership in dollar market capitalization. Size is the log of market capitalization of the firm. Leverage refers to the leverage ratio, calculated as the firm's total debt divided by the market value of total assets. Cash is the ratio of cash and short-term investments to total assets. Heteroskedasticity-consistent *t*-statistics clustered at the firm level are reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	(1) IO	(2) IO	(3) Log(IO_USD)	(4) Log(IO_USD)
Green*Post	0.056*	0.079**	0.099***	0.097***
	(2.03)	(2.71)	(5.40)	(4.71)
Post	0.093***	0.085***	0.0843***	0.0973***
	(3.02)	(3.19)	(9.52)	(4.52)
Green	-0.034***	-0.020***	-0.0409***	-0.0134*
	(-10.85)	(-3.95)	(-8.73)	(-1.85)
Size		-0.018***		0.384***
		(-5.91)		(26.68)
Leverage		0.144***		1.981***
		(5.52)		(7.45)
ROA		0.512***		7.435***
		(4.36)		(11.57)
Cash		-0.127**		-2.299***
		(-2.22)		(-5.03)
Profitability		0.116***		0.006
		(10.59)		(0.05)
Dividend		-0.251***		-0.565**
		(-5.79)		(-2.50)
Country FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Robust	YES	YES	YES	YES
R-squared	0.042	0.195	0.100	0.267
Observations	7179	6739	7179	6739

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Table 6. Institutional ownership: Foreign vs. domestic investors

This table presents the results of foreign and domestic institutional investors' ownership change after firms have successfully labeled their bonds green. Following Ferreira, Miguel, and Pedro Matos, 2008, the data range from 2000 to 2017. Institutional ownership is defined as total institutional ownership in percentage of market capitalization. Independent institutions include mutual funds and independent investment advisers. Size is the log of market capitalization of the firm. Leverage refers to the leverage ratio, calculated as the firm's total debt divided by the market value of total assets. Cash is the ratio of cash and short-term investments to total assets. Heteroskedasticity-consistent *t*-statistics clustered at the firm level are reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	(1)	(2)	(3)	(4)
	io_dom	io_dom_indep	io_for	io_for_indep
Green*Post	0.085***	0.076**	-0.005	-0.019***
	(3.20)	(2.88)	(-1.65)	(-3.52)
Post	0.097***	0.094***	-0.012*	-0.017**
	(3.76)	(3.61)	(-1.76)	(-2.34)
Green	-0.007	-0.007	-0.013***	-0.024***
	(-1.55)	(-1.69)	(-7.31)	(-10.46)
Size	-0.018***	-0.018***	-0.000	0.005***
	(-7.57)	(-8.01)	(-0.04)	(2.92)
leverage	0.186***	0.194***	-0.041**	-0.053*
	(8.65)	(9.90)	(-2.53)	(-2.06)
ROA	0.258**	0.285**	0.255***	0.229***
	(2.22)	(2.68)	(9.80)	(6.29)
Cash	-0.222***	-0.192***	0.095***	0.059***
	(-3.78)	(-3.43)	(4.86)	(3.35)
Profitability	0.095***	0.070***	0.022***	0.013**
	(7.06)	(6.02)	(3.46)	(2.36)
Dividend	-0.353***	-0.342***	0.102***	0.017
	(-9.92)	(-10.24)	(7.32)	(1.23)
Country FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Robust	YES	YES	YES	YES
R-Squared	0.197	0.201	0.183	0.562
Observations	6739	6739	6739	6739

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Table 7. Stock liquidity before and after issuing green bond

Amihud illiquidity measure is calculated using absolute stock return divided by dollar trading volume on that day. We take a monthly average of all the available trading days in that calendar month. The higher the Amihud measure, the less liquid the stock. The quoted spread is calculated using bid price minus ask price divided by the average of bid price and ask price. The larger the quoted spread, the less liquid the stock. Because our liquidity measure is highly skewed, we take the logarithm. Heteroskedasticity-consistent *t*-statistics clustered at the firm level are reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	(1)	(2)	(3)	(4)
	Quoted Spread	Quoted Spread	Amihud	Amihud
Post	-0.0546***	-0.0830***	-0.152***	-0.127***
	(-2.964)	(-4.457)	(-2.614)	(-3.234)
Price		-0.00453***		-0.00673*
		(-3.136)		(-1.694)
Volume(log)		-0.112***		-0.610***
		(-6.364)		(-6.776)
Volatility		0.0414***		0.0987***
		(2.783)		(4.130)
Size		-0.135***		-0.304***
		(-2.644)		(-2.769)
Firm FE	YES	YES	YES	YES
Year FE	NO	YES	NO	YES
R-squared	0.03	0.48	0.015	0.461
Observations	2,914	2,792	2,914	2,792

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Table 8. Stock liquidity before and after green bond issuance: Matching sample analysis

Amihud illiquidity measure is calculated using absolute stock return divided by dollar trading volume on that day. We take a monthly average of all the available trading days in that calendar month. The higher the Amihud measure, the less liquid the stock. The quoted spread is calculated using bid price minus ask price divided by the average of bid price and ask price. The larger the quoted spread, the less liquid the stock. Because our liquidity measure is highly skewed, we take the logarithm. We match our sample according to size, market to book and previous year liquidity to control for the trend. We further restrict our matching sample to have a normal bond issued in the same year. Heteroskedasticity-consistent *t*-statistics clustered at the firm level are reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	Quoted Spread		Amihud	
	(1)	(2)	(3)	(4)
	All sample	All sample	All sample	All sample
Green*Post	-0.0672*	-0.01274**	-0.0110**	-0.022**
	(-1.968)	(-2.09)	(-1.962)	(-2.44)
Post	0.0122	-0.0141***	-0.0476***	-0.0119***
	(-0.37)	(-3.35)	(-12.57)	(-9.26)
Green	0.1091	-0.0835	0.0863	-0.0633
	(-1.345)	(-0.26)	(-0.482)	(-0.364)
Control	NO	YES	NO	YES
Year FE	YES	YES	YES	YES
Country FE	YES	YES	YES	YES
R-squared	0.012	0.246	0.024	0.483
Observations	5,824	5,442	5,824	5,442

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Table 9. Stock liquidity before and after green bond issuance: Initial ESG condition

The Amihud illiquidity measure is calculated using absolute stock return divided by dollar trading volume on that day. We take a monthly average of all the available trading days in that calendar month. The higher the Amihud measure, the less liquid the stock. The quoted spread is calculated using bid price minus ask price divided by the average of bid price and ask price. The larger the quoted spread, the less liquid the stock. Because our liquidity measure is highly skewed, we take the logarithm. ESG coverage is defined as this firm being covered by the RepRisk ESG database. We match our sample according to size, market to book and previous year liquidity to control for the trend. We further restrict our matching sample to have a normal bond issued in the same year. Heteroskedasticity-consistent *t*-statistics clustered at the firm level are reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	(1)	(2)	(3)	(4)
	Quoted_Spread	Quoted_Spread	Amihud	Amihud
Green_Post_ESG coverage	0.00345***	0.00120	0.00291*	0.00600**
	(2.849)	(0.605)	(1.818)	(2.457)
Green	0.0730	0.0332	0.0103	-0.0106***
	(0.0906)	(1.016)	(0.130)	(-5.206)
Post	-0.0181**	0.0962	-0.0300***	0.01252
	(-2.349)	(1.244)	(-2.948)	(1.312)
ESG	-00.717	0.0857	-0.0219***	-0.0103**
	(-0.849)	(1.179)	(-2.624)	(-2.218)
Green_Post	-0.0195*	-0.0115	-0.0139	-0.0190*
	(-1.696)	(-1.415)	(-0.923)	(-1.959)
Size		-3.09		-2.16***
		(-1.373)		(-7.212)
Profitability		0.177		0.163
		(1.282)		(0.939)
Leverage		0.699***		0.400*
		(4.200)		(1.928)
Dividend		0.0518		-0.408***
		(0.780)		(-5.155)
Country FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
R-squared	0.031	0.063	0.122	0.158
Observations	5,824	5,442	5,824	5,442

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Internet Appendix for

"Do Shareholders Benefit from Green Bonds?"

(not for publication)

This Internet Appendix contains additional robustness test and baseline sample details for the paper "Do Shareholders benefit from Green Bonds?". Figures IA.1 and IA.2 show two different types of green bonds' prospectuses. Tables IA.3 and IA.4 show the results not reported in the main table due to space constraints. Table IA.5 provides detailed information about all the public green bond issuers we used in all our tests in the paper.

The tables in the appendix are as follows:

- Table IA.1: displays one green bond prospectus issued by a corporation: Apple Inc.
- Table IA.2: displays one green bond prospectus issued by a bank: the Bank of America.
- Table IA.3: illustrates the decomposition of increased institutional ownership due to the issuance of green bonds.
- Table IA.4: displays the robustness of Table 8 in the paper to prove that our stock liquidity improvement results are robust to changes in window.
- Table IA.5: displays detailed information about public green bond issuers, including their announcement date, coupon, maturity, issue amount, currency and home country.

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Internet Appendix IA.1

Excerpt of Apple Green Bond Prospectus (Corporate Issuer)

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Prospectus Supplement
(To Prospectus dated April 29, 2013)

$12,000,000,000

Apple Inc.

$500,000,000 Floating Rate Notes due 2019
$500,000,000 Floating Rate Notes due 2021
$500,000,000 1.300% Notes due 2018
$1,000,000,000 1.700% Notes due 2019
$2,250,000,000 2.250% Notes due 2021
$1,500,000,000 2.850% Notes due 2023
$2,000,000,000 3.250% Notes due 2026
$1,250,000,000 4.500% Notes due 2036
$2,500,000,000 4.650% Notes due 2046

We are offering $500,000,000 of our Floating Rate Notes due 2019 (the "2019 Floating Rate Notes"), $500,000,000 of our Floating Rate Notes due 2021 (the "2021 Floating Rate Notes" and, together with the 2019 Floating Rate Notes, the "floating rate notes"), $500,000,000 of our 1.300% Notes due 2018 (the "2018 Fixed Rate Notes"), $1,000,000,000 of our 1.700% Notes due 2019 (the "2019 Fixed Rate Notes"), $2,250,000,000 of our 2.250% Notes due 2021 (the "2021 Fixed Rate Notes"), $1,500,000,000 of our 2.850% Notes due 2023 (the "2023 Fixed Rate Notes"), $2,000,000,000 of our 3.250% Notes due 2026 (the "2026 Fixed Rate Notes") , $1,250,000,000 of our 4.500% Notes due 2036 (the "2036 Fixed Rate Notes") and $2,500,000,000 of our 4.650% Notes due 2046 (the "2046 Fixed Rate Notes" and, together with the 2018 Fixed Rate Notes, the 2019 Fixed Rate Notes, the 2021 Fixed Rate Notes, the 2023 Fixed Rate Notes, the 2026 Fixed Rate Notes and the 2036 Fixed Rate Notes, the "fixed rate notes"). We refer to the floating rate notes and the fixed rate notes collectively as the "notes."

The 2019 Floating Rate Notes will bear interest at a floating rate equal to three-month LIBOR plus 0.820% and mature on February 22, 2019. The 2021 Floating Rate Notes will bear interest at a floating rate equal to three-month LIBOR plus 1.130% and mature on February 23, 2021. We will pay interest on the floating rate notes quarterly in arrears on February 23, May 23, August 23 and November 23 of each year, beginning on May 23, 2016.

We will pay interest on the fixed rate notes semi-annually in arrears on February 23 and August 23 of each year, beginning on August 23, 2016. The 2018 Fixed Rate Notes will mature on February 23, 2018, the 2019 Fixed Rate Notes will mature on February 22, 2019, the 2021 Fixed Rate Notes will mature on February 23, 2021, the 2023 Fixed Rate Notes will mature on February 23, 2023, the 2026 Fixed Rate Notes will mature on February 23, 2026, the 2036 Fixed Rate Notes will mature on February 23, 2036 and the 2046 Fixed Rate Notes will mature on February 23, 2046.

We may redeem the fixed rate notes in whole or in part at any time or from time to time at the redemption prices described under the heading "Description of the Notes—Optional Redemption" in this prospectus supplement. The floating rate notes may not be redeemed before maturity. The notes will be issued only in minimum denominations of $2,000 and integral multiples of $1,000 in excess thereof.

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USE OF PROCEEDS

Use of Proceeds for Floating Rate Notes and 2018, 2019, 2021, 2026, 2036, and 2046 Fixed Rate Notes

We estimate the net proceeds from sales of the notes (other than the 2023 Fixed Rate Notes) will be approximately $10.44 billion, after deducting underwriting discounts and the *pro rata* offering expenses for such notes. We intend to use such net proceeds for general corporate purposes, including repurchases of our common stock and payment of dividends under our program to return capital to shareholders, funding for working capital, capital expenditures, acquisitions and repayment of debt.

We may temporarily invest funds that are not immediately needed for these purposes in short-term investments, including cash, cash equivalents and/or marketable securities.

Use of Proceeds for 2023 Fixed Rate Notes

We estimate the net proceeds from sales of the 2023 Fixed Rate Notes will be approximately $1.49 billion, after deducting underwriting discounts and the *pro rata* offering expenses for the 2023 Fixed Rate Notes. An amount equal to such net proceeds will be allocated for investments in one or more Eligible Projects (as defined below), including new and ongoing development of such Eligible Projects. Eligible Projects may include projects of our subsidiaries that meet the Eligibility Criteria set forth below.

We plan to identify Eligible Projects that fall within three environmental priorities where we believe we can make the most environmentally positive impact:

1. Reduce our impact on climate change by using renewable energy sources and driving energy efficiency in our facilities, products and supply chain.
2. Pioneer the use of greener materials in our products and processes.
3. Conserve precious resources.

"*Eligible Projects*" means projects that meet the Eligibility Criteria (as defined below).

"*Eligibility Criteria*" means any of the following:

- expenditures related to new and ongoing renewable energy projects, such as solar and wind projects, or associated energy storage solutions,
- expenditures related to projects that have received within the last three years, or are expected to receive, certification of LEED Gold or Platinum or BREEAM Very Good, Excellent, or Outstanding "green building" standards,
- expenditures related to the implementation of environmental design elements for new or ongoing building developments, such as high performance mechanical systems, natural ventilation, on-site renewable energy, and high performance lighting systems,
- expenditures related to energy efficiency projects and technologies at our corporate facilities, such as heating, ventilation and air

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Internet Appendix IA. 2

Excerpt of Bank of America Green Bond Prospectus (Financial Institution Issuer)

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Filed Pursuant to Rule 424(b)(3)
Registration No. 333-202354
Prospectus Addendum
(To Prospectus Dated May 1, 2015 and Prospectus Supplement Dated October 17, 2016)
November 4, 2016

Bank of America

As part of our Bank of America Corporation Medium-Term Notes, Series L, program, we may offer from time to time notes where the proceeds will be used to further our consolidated company's environmental business initiative.

In 2012, our consolidated company announced a new 10-year $50 billion environmental business initiative to help address climate change, reduce demands on natural resources and advance lower-carbon economic solutions. We have since increased this commitment to $125 billion in low carbon business by 2025. As part of this initiative, we are focusing on energy efficiency, renewable energy and energy infrastructure, advanced transportation, waste and water, through lending, investing and facilitating capital, providing advice and developing solutions for clients around the world.

We may offer from time to time our Medium-Term Notes, Series L, where we specify that the proceeds from such offering will be used in furtherance of this initiative, in particular to finance renewable energy projects. The specific terms of any notes that we offer in conjunction with our company's environmental business initiative will be determined before each sale and will be described in a separate pricing supplement. The terms of any such notes will not be tied to any specific environmental initiative or the performance or success of any environmental initiative.

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USE OF PROCEEDS

Eligible Green Projects

An amount equal to the net proceeds from any sale of notes in furtherance of our environmental business initiative will be used to fund renewable energy projects ("Eligible Green Projects"), in whole or in part, as defined by Bank of America internal investment criteria. Renewable energy projects include financing of, or investments in, equipment and systems which facilitate the use of energy from renewable sources, such as solar, wind, and geothermal energy. The specific use of proceeds of any note issuance will be described in the applicable pricing supplement.

Project Evaluation and Selection

The Eligible Green Projects are identified and selected via a process that involves participants from various functional areas including our Global Environmental Group, our Corporate Treasury group and our Global Banking group.

Our Global Environmental Group evaluates and determines project eligibility according to the criteria indicated above. The list of projects compiled by the Global Environmental Group is reviewed and approved by our Corporate Treasury group and our Global Banking group.

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Internet Appendix IA. 3
Different Types of Institutional Ownership

This table presents the breakdown of institutional investors' ownership change after firms, such as insurance companies, investment advisers, pension funds, hedge funds and banks, have successfully labeled their bonds green. Following Ferreira, Miguel, and Pedro Matos, 2008, the data range from 2000 to 2017. Institutional ownership is defined as total institutional ownership in percentage of market capitalization. Size is the log of market capitalization of the firm. Leverage refers to the leverage ratio, calculated as the firm's total debt divided by the market value of total assets. Cash is the ratio of cash and short-term investments to total assets. Heteroskedasticity-consistent standard errors clustered at the firm level are reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	(1) Insurance	(2) Investment adviser	(3) Pension Fund	(4) Hedge Fund	(5) Bank
Green*Post	0.0142**	0.0210***	0.0681***	-0.0138***	-0.0435***
	(0.00641)	(0.00767)	(0.0206)	(0.00187)	(0.00519)
Size	0.116***	0.0171***	-0.217**	-0.558***	-0.0382
	(0.0338)	(0.00405)	(0.109)	(0.0986)	(0.0274)
Leverage	0.596***	-0.170***	0.591	0.0296***	0.0312***
	(0.138)	(0.0165)	(0.442)	(0.00401)	(0.0111)
ROA	0.0123***	0.0198	-0.0177*	0.653	-0.0434
	(0.00319)	(0.0382)	(0.0102)	(0.930)	(2.58)
Cash	0.0171***	0.0955***	-0.765	0.821	0.585***
	(0.00184)	(0.0221)	(0.592)	(0.538)	(0.149)
R&D	0.924	-0.662***	-0.0631	0.129***	0.0683
	(1.68)	(0.201)	(0.0540)	(0.0490)	(0.136)
Sale	0.107	0.0460***	0.880**	-0.625*	3.70
	(0.114)	(0.0136)	(0.366)	(0.332)	(9.22)
Dividend	0.348***	-0.0153	-0.579*	0.0137***	0.282***
	(0.0958)	(0.0115)	(0.308)	(0.00279)	(0.0776)
Observations	1,961	1,961	1,961	1,961	1,961
R-squared	0.608	0.901	0.859	0.796	0.547
Firm FE	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES
Robust	YES	YES	YES	YES	YES

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Internet Appendix IA.4
Liquidity Changes around Green Bond Issuance: Different Time Horizon

Amihud illiquidity measure is calculated using absolute stock return divided by dollar trading volume in that day. We take a monthly average of all the available trading days in that calendar month. The quoted spread is calculated using bid price minus ask price divided by the average of bid price and ask price. The higher the Amihud measure, the less liquid the stock. The larger the quoted spread, the less liquid the stock. Because our liquidity measure is highly skewed, we take the logarithm. (Robust standard errors in parentheses*** p<0.01, ** p<0.05, * p<0.1.)

Liquidity	Amihud			Quoted Spread		
	(1)	(2)	(3)	(4)	(5)	(6)
	[-6m,6m]	[-8m,8m]	[-10m,10m]	[-6m,6m]	[-8m,8m]	[-10m,10m]
Green*Post	-0.0921**	-0.0983*	-0.112**	-0.0466**	-0.0773***	-0.0758***
	(0.0462)	(0.0497)	(0.0481)	(0.0226)	(0.0214)	(0.0200)
Price	-0.0155*	-0.0107	-0.00876	-0.00573**	-0.00586***	-0.00501***
	(0.00784)	(0.00712)	(0.00591)	(0.00276)	(0.00205)	(0.00168)
Volume	-660.2	-1,422	-2,152	1,078	937.1	783.7
	(835.1)	(1,488)	(1,924)	(683.3)	(664.8)	(606.2)
Volatility	0.0335**	0.0359**	0.0521***	0.0287	0.0327	0.0318*
	(0.0162)	(0.0167)	(0.0167)	(0.0230)	(0.0221)	(0.0178)
Size	-0.841***	-0.895***	-0.916***	-0.147*	-0.0828	-0.166***
	(0.160)	(0.135)	(0.133)	(0.0775)	(0.0643)	(0.0559)
Constant	-6.107***	-5.740***	-5.608***	-4.942***	-5.502***	-4.798***
	(1.381)	(1.171)	(1.151)	(0.679)	(0.562)	(0.487)
Observations	1,401	1,785	2,155	1,384	1,764	2,130
R-squared	0.110	0.118	0.136	0.017	0.021	0.025
Firm FE	YES	YES	YES	YES	YES	YES

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Internet Appendix IA.5: List of Public Green Bond Issuers

Number	Issuer Name	Announce	Amount (Million)	Currency	Country	Coupon (%)	Maturity (Y)
1	Nedbank	8/17/2012	4000	ZAR	ZAF	.	
2	Regency Centers Corp	10/22/2012	300	USD	USA	3.75	6.5
3	Credit Agricole CIB	1/29/2013	7	BRL	FRA	6	3.5
4	Bank of America	11/18/2013	500	USD	USA	1.35	3
5	NRW Bank	11/19/2013	250	EUR	DEU	0.75	4
6	EDF	11/20/2013	1400	EUR	FRA	2.25	7.4
7	Unibail-Rodamco	2/19/2014	750	EUR	FRA	2.5	10
8	Unilever	3/19/2014	250	GBP	GBR	2	4.7
9	Toyota	3/19/2014	490	USD	USA	1.3	3.9
10	ESSITY AB	3/25/2014	1000	SEK	SWE	0.217	5
11	TD Bank	3/28/2014	500	CAD	CAN	1.824	3
12	Skanska	4/1/2014	850	SEK	SWE	0.465	5
13	Iberdrola	4/8/2014	750	EUR	NLD	2.5	8.5
14	Arise	4/16/2014	1100	SEK	SWE	3	5
15	Engie	5/12/2014	1200	EUR	FRA	1.375	6
16	Vornado Realty Trust	6/9/2014	450	USD	USA	2.5	5.1
17	Hera SpA	6/26/2014	500	EUR	ITA	2.375	10
18	Advanced Semiconductor Engineering	7/17/2014	300	USD	TWN	2.125	3
19	Abengoa Greenfield	9/24/2014	270	EUR	ESP	0	12.5
20	Stockland Trust Management Ltd	10/23/2014	300	EUR	AUS	1.5	7
21	Verbund AG	11/13/2014	500	EUR	AUT	1.5	10
22	DNB Bank	2/10/2015	1000	NOK	NOR	1.35	5
23	Vestas	3/4/2015	500	EUR	DNK	2.75	7
24	SolarCity	3/16/2015	2.8	USD	USA	3.6	5
25	Wallenstam	3/18/2015	500	SEK	SWE	0.594	4
26	Berlin Hyp	4/27/2015	500	EUR	DEU	0.125	7
27	ANZ Bank	5/27/2015	600	AUD	AUS	3.25	5
28	BRF SA	5/29/2015	500	EUR	BRA	2.75	7
29	Morgan Stanley	6/3/2015	500	USD	USA	2.2	3.5
30	Digital Realty Trust	6/18/2015	500	USD	USA	3.95	7
31	NRG Yield	6/22/2015	500	USD	USA	5.375	10
32	Xinjiang Goldwind	7/16/2015	300	USD	HKG	2.5	3
33	NWB Bank	8/25/2015	1000	EUR	NLD	1	10
34	CLP Wind Farms IND	9/15/2015	2000	INR	IND	9.15	3.6
35	LM Group Holding	9/25/2015	480	NOK	DNK	0	0
36	Schneider Electric SE	10/7/2015	300	EUR	FRA	1.841	10
37	Agricultural Bank of CHN	10/13/2015	400	USD	CHN	2.125	3

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38	Sumitomo Mitsui Banking Corp	10/14/2015	500	USD	JPN	2.45	5
39	Southern Power Company	11/12/2015	500	USD	USA	4.15	10
40	ING	11/17/2015	500	EUR	NLD	0.75	5
41	Societe Generale	11/18/2015	500	EUR	FRA	0.75	5
42	IDBI	11/23/2015	350	USD	IND	4.25	5
43	HSBC	11/25/2015	500	EUR	FRA	0.625	5
44	SPD Bank	1/20/2016	20000	CNY	CHN	2.95	3
45	Industrial Bank Co, Ltd.	1/25/2016	10000	CNY	CHN	2.95	3
46	Apple, Inc.	2/16/2016	1500	USD	USA	2.85	7
47	Georgia Power Company	3/2/2016	330	USD	USA	3.25	10.1
48	Hyundai Capital Services, Inc.	3/7/2016	500	USD	KOR	2.875	5
49	Century Concord Wind Power	4/1/2016	200	CNY	CHN	6.2	3
50	BAIC Motors	4/14/2016	2500	CNY	CHN	3.45	7
51	Turkiye Sinai Kalkinma Bankasi AS	5/12/2016	300	USD	TUR	4.875	5
52	Axis Bank	5/23/2016	500	USD	IND	2.875	5
53	Westpac Banking Corp	5/25/2016	500	AUD	AUS	3.1	5
54	Westar Energy, Inc.	6/13/2016	350	USD	USA	2.55	10
55	Huaneng Renewables	7/7/2016	1100	CNY	CHN	2.95	5
56	Suzano Papel e Celulose	7/7/2016	500	USD	BRA	5.75	10
57	Link REIT	7/14/2016	500	USD	HKG	2.875	10
58	Bajaj Finance, Ltd.	7/14/2016	100	INR	IND	8.55	5
59	Beijing Enterprises Water Group, Ltd.	7/28/2016	700	CNY	CHN	3.25	8
60	NTPC	8/3/2016	20000	INR	IND	7.375	5
61	CHN Three Gorges Corp	8/25/2016	3500	CNY	CHN	2.92	3
62	Mitsubishi UFG	9/6/2016	500	USD	JPN	2.527	7
63	Nomura Research Institute	9/9/2016	10000	JPY	JPN	0.25	10
64	CHN Datang	9/13/2016	1000	CNY	CHN	3.5	5
65	Entra ASA	9/16/2016	1000	NOK	NOR	1.82	7
66	CHN Energy Conservation & Environmental Protection	9/21/2016	1500	CNY	CHN	3.55	10
67	Castellum	9/26/2016	350	SEK	SWE	1.875	5
68	CPFL Energias Renováveis	9/28/2016	200	BRL	BRA	8.37	5
69	Aggregated Micro Power Infrastructure 2 Plc	10/12/2016	52	GBP	GBR	8	20
70	Modern Land CHN Co Ltd	10/13/2016	500	USD	CHN	6.875	3
71	Beijing SPC Environment Protection Tech Co Ltd	10/21/2016	1090	CNY	CHN	3.7	5
72	MTR	10/24/2016	600	USD	HKG	2.5	10
73	Acciona	11/4/2016	22	EUR	ESP	4.25	15
74	Bank of Communications	11/15/2016	10000	CNY	CHN	2.94	3

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75	BNP Paribas	11/24/2016	500	EUR	FRA	0.5	5.5
76	BanCOL	12/5/2016	350000	COP	COL	IBR+220bp	7
77	Green Bancorp, Inc.	12/8/2016	34	USD	USA	8.5	10
78	Enel	1/9/2017	1300	EUR	ITA	1	7.7
79	Fibria	1/11/2017	700	USD	BRA	5.5	10
80	Panda Green Energy Group, Ltd.	1/18/2017	350	USD	CHN	8.25	3
81	Pattern Energy Group, Inc.	1/20/2017	350	USD	USA	5.875	7
82	CHN Huarong Financial Leasing Co, Ltd.	2/9/2017	1000	CNY	CHN	4.45	3
83	Chongqing Longhu Development Co, Ltd.	2/9/2017	1000	CNY	CHN	4.67	7
84	SEB	2/10/2017	500	EUR	SWE	0.3	5
85	FlexiGroup	2/17/2017	265	AUD	AUS	2.88	
86	Covanta Holding Corp	3/2/2017	400	USD	USA	5.875	8.3
87	Dongjiang Environmental Co, Ltd.	3/8/2017	600	CNY	CHN	4.9	3
88	Atrium Ljungberg	3/13/2017	200	SEK	SWE	1.619	5
89	Commonwealth Bank of AUS	3/20/2017	200	AUD	AUS	2.625	5
90	National Bank of Abu Dhabi	3/27/2017	590	USD	ARE	3	5
91	Harbin Bank	3/31/2017	2000	CNY	CHN	4.79	3
92	CDL Properties	4/6/2017	100	SGD	SGP	1.98	2
93	Bank of Beijing	4/12/2017	12000	CNY	CHN	4.3	3
94	Bank of Nanjing Co, Ltd.	4/20/2017	1000	CNY	CHN	4.6	5
95	QBE Insurance	4/20/2017	300	USD	AUS	3	5.5
96	CTBC Commercial Bank	5/9/2017	1000	TWD	TWN	0.83	3
97	Longyuan Power	5/11/2017	2000	CNY	CHN	4.9	5
98	Bank Sinopac	5/12/2017	45	USD	TWN	0	30
99	E Sun Commercial Bank	5/16/2017	60	USD	TWN	0	30
100	Yunnan Water Investment Co, Ltd.	5/19/2017	550	CNY	CHN	6.3	10
101	Bank Zachodni WBK	5/25/2017	137	EUR	POL	.	
102	CHN Huadian Corp	6/6/2017	1500	CNY	CHN	4.8	5
103	Nordea Bank	6/16/2017	500	EUR	SWE	0.3	5
104	Intesa Sanpaolo SpA	6/16/2017	500	EUR	ITA	0.875	5
105	Helvetia Environnement Groupe	6/21/2017	75	CHF	CHE	3.625	5
106	Grupo Rotoplas	6/26/2017	1400	MXN	MEX	8.65	10
107	L&T Infrastructure Finance Company, Ltd.	6/29/2017	6700	INR	IND	7.59	7.4
108	Rural Electrification Corp	6/29/2017	450	USD	IND	3.875	10
109	Brookfield Renewable Partners	7/12/2017	475	USD	CAN	4.4	15
110	Terna Energy Finance SA	7/14/2017	60	EUR	GRC	3.85	5
111	DBS Group	7/17/2017	500	USD	SGP	1.93444	5

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112	Tesla, Inc.	8/7/2017	1800	USD	USA	5.3	8
113	Guangzhou Development Group, Inc.	8/29/2017	2400	CNY	CHN	4.94	5
114	Inner Mongolia M-Grass Ecology and Environment Group Co, Ltd.	8/30/2017	250	CNY	CHN	5.5	3
115	SSE PLC	8/30/2017	600	EUR	GBR	0.875	8
116	Bank of Zhengzhou Co, Ltd.	8/31/2017	3000	CNY	CHN	4.7	3
117	Huishang Bank Corp, Ltd.	9/1/2017	1000	CNY	CHN	4.49	3
118	ICADE	9/4/2017	600	EUR	FRA	1.5	10
119	CECEP Wind-Power Corp	9/5/2017	300	CNY	CHN	4.83	5
120	Sumitomo Mitsui Financial Group, Inc.	10/4/2017	500	EUR	JPN	0.934	7
121	Mizuho Financial Group, Inc.	10/10/2017	500	EUR	JPN	0.956	7
122	Jiangsu Zhongli Group Co, Ltd.	10/24/2017	100	CNY	CHN	6.5	3
123	Taiwan Business Bank	10/26/2017	1000	TWD	TWN	0.7	3
124	Oriental Energy Co, Ltd.	10/31/2017	600	CNY	CHN	6.5	3
125	Barclays PLC	11/6/2017	500	EUR	GBR	0.625	6
126	Manulife Financial Corp	11/14/2017	500	SGD	CAN	3	12
127	Avangrid, Inc.	11/16/2017	600	USD	USA	3.15	7
128	Orsted A/S	11/16/2017	500	EUR	DNK	2.25	1000
129	Shenzhen Energy Group Co, Ltd.	11/20/2017	1000	CNY	CHN	5.25	5
130	Power Finance Corp, Ltd.	11/28/2017	400	USD	IND	3.75	10
131	Shandong Lipeng Co, Ltd.	11/29/2017	250	CNY	CHN	6.5	5
132	Toda Corp	12/8/2017	10000	JPY	JPN	0.27	5

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